Supplementary Material: Literature Selection and Comparative Evidence

This supplement accompanies *Wheeled Mobile Robot Autonomy Under Degradation: A Survey of Methods and Evaluation*. It provides the selection and assessment details summarized in Section III, selected numerical comparisons discussed in the technical synthesis, and the tables of representative implementations referenced in Sections IV–VIII. Counts describe the selected literature and do not estimate research-topic prevalence across the field.

S1. Search Coverage and Selection

The initial search emphasized 2021–2026 publications across 17 journals and 102 journal–year combinations. It yielded 58,583 merged records, of which 43,086 had sufficient metadata for title/abstract screening. Language-model-assisted screening used preset topic allocations and stopped after 45 acceptances, leaving 43,041 metadata-complete records unassessed. Selection emphasized mechanisms relevant to compact wheeled platforms. The screening order and complete rejection history are unavailable. These missing records and the stopping rule limit the representativeness of the sample.

The archived screening decisions of the later candidate queues comprise 220 records, of which 150 accept, 40 reject, and 30 defer a candidate. Every record names Codex as reviewer and execution host. All but four state one of three title-and-abstract relevance policies. The read scope is recorded for 85 records: title and abstract for 74, title only for 7, metadata only for 3, and the abstract with the authors' platform description for one. Hard-gate results are recorded for 119 records, of which 94 passed. Gate observations are recorded for 43 more. An identity check of DOI, venue, year, publication type, and main-conference status is recorded for 57 records. No record states the language model or its version. The initial screening left only the aggregate counts given above. The public data list the follow-up screening of 79 candidates with reviewer and read scope.

Additional coverage comprised 5 complementary publications, a 60-work thematic sample, 15 studies selected for the four integration questions developed in the survey, 20 IEEE publications concerning motion priors, recovery, shared-state diagnosis, and reproducibility, a further 48 targeted works, 36 recent IEEE publications, and 16 coverage additions. The 48-work stage covers degeneracy and map validity, constrained motion and terrain risk, multimodal navigation, communication-aware collaboration, and recovery or online learning. The recent IEEE stage selected six 2025–2026 IEEE journal or ICRA papers for each of the six themes after an exact-DOI check against the works already selected. The selection table in Section III of the main paper summarizes these additions. The thematic sample contained 45 journal articles and 15 conference papers, with ten works under each of six themes: perception/localization, mapping/exploration, planning/control, multi-robot coordination, semantic navigation, and reliability/deployment. Searches combined technical, scenario, and mechanism terms with citation follow-up. Screening-based selection was completed by November 3, 2026, leaving coverage of the 2026 calendar year incomplete.

Table S1 gives venue type and publisher by stage. The targeted and recent IEEE stages searched IEEE venues only, which makes all 56 of their works IEEE publications. The records of these stages state no reason for the restriction. Outside them, IEEE publications form 64% of the works. Without the two stages, the headline share of Section S6 is 33% instead of 33% (Table S10).

The final stage, on November 4, 2026, added 16 works from 2021–2024 that an internal review of an earlier draft named as missing. They comprise foundation-model and open-vocabulary navigation [1]–[5], uncertainty-aware terrain risk [6], [7], outlier-robust multi-robot SLAM [8], [9], DARPA Subterranean Challenge team systems [10]–[12], and datasets for degraded SLAM, off-road perception and dynamics, and embodied navigation [13]–[16]. They were not identified by a systematic search. The review also named VLMaps, whose 2025 journal extension is already in the corpus [17], NeBula, which exists only as a preprint, and further works that this stage did not acquire. Without these additions, 77 of the 228 purposively selected technical publications (34%) report a task outcome under a named degradation condition in the same experiment, against 81 of 244 (33%) with them.

A. Journal and Conference Sources

The journal searches covered Automatica, Autonomous Robots, Control Engineering Practice, IEEE Robotics and Automation Letters, IEEE Transactions on Automatic Control, IEEE Transactions on Automation Science and Engineering, IEEE Transactions on Cybernetics, IEEE Transactions on Image Processing, IEEE Transactions on Intelligent Transportation Systems, IEEE Transactions on Intelligent Vehicles, IEEE Transactions on Robotics, Information Fusion, Journal of Field Robotics, Pattern Recognition, Robotics and Autonomous Systems, Science Robotics, and The International Journal of Robotics Research. The 2025–2026 searches covered 34 journal–year combinations. Completion of a search provider's enumeration does not establish omission-free retrieval of the publisher's catalog.

Formal CoRL volumes for 2024 and 2025 and RSS indexes for 2024–2026 were consulted. ICRA/IROS papers were assessed individually, without exhaustive enumeration of their proceedings. Complete enumeration of a formal CoRL 2026

TABLE S1
VENUE TYPE AND PUBLISHER OF THE SELECTED WORKS BY SELECTION STAGE. THE PUBLISHER FOLLOWS THE DOI PREFIX, OR THE OFFICIAL URL HOST FOR WORKS WITHOUT A DOI.

Stage	Works	Journal	Conf.	IEEE	Elsevier	Wiley	SAGE	Springer	PMLR	Other	IEEE %
Initial sample	45	45	0	38	3	3	1	0	0	0	84
Complementary publications	5	5	0	3	0	0	1	0	0	1	60
Thematic sample	60	45	15	42	6	1	3	1	2	5	70
Integration questions	15	13	2	8	1	1	1	2	2	0	53
Targeted IEEE coverage	20	17	3	20	0	0	0	0	0	0	100
Further targeted coverage	48	36	12	21	2	7	6	2	10	0	44
Recent IEEE coverage	36	34	2	36	0	0	0	0	0	0	100
Coverage additions	16	6	10	9	0	0	0	0	2	5	56
All stages	245	201	44	177	12	12	12	5	16	11	72

Other: AAAS, Robotics: Science and Systems, NeurIPS Datasets and Benchmarks, Field Robotics and Tsinghua University Press. The context review is included in its stage.

TABLE S2
SELECTED RESULTS AND WITHIN-STUDY COMPARISONS. TABLE AND FIGURE NUMBERS IN THE FIRST COLUMN REFER TO THE CITED PAPERS. EACH ROW RETAINS ITS OWN METRIC AND PROTOCOL.

Study / source	Protocol and measure	Reported result	Interpretation
LODESTAR [18] Tables III and V	SubT-MRS Long Corridor replay: absolute pose error (APE). Desktop computation averaged across benchmark recordings.	APE 1.386 m versus 3.113 m for FAST-LIO2; benchmark-average computation 28.69 versus 21.04 ms.	Selective anchoring and point selection improve this sequence at additional computation cost; no LiDAR-blackout or closed-loop test.
FIT-SLAM 2 [19] Tables 2 and 4	Exploration with simulator poses and common control; completion time, plus Fisher-information timing analysis.	Cave: 880 versus 1118 s for FAEL; Corridor: 1809 versus 1706 s. Planning: 20.8 ms without and 63.0 ms with Fisher information.	The ordering changes with the scene. Planner benchmarks and the SLAM/localizability ablation answer different questions.
Safe Horizon MPC [20] Table 2	100 Gaussian eight-pedestrian simulations; common solver/cost and matched joint-risk specification 0.05; duration mean (SD).	16.0 (1.9) s versus 18.4 (2.4) s for CC-MPC. Maximum estimated plan collision probabilities: 0.0092 versus 0.0019.	Faster traversal accompanies less conservative plan risk under a known motion distribution; these are not empirical mission collision rates.
LeLaN [21] Tables 2 and 3	Visible-target navigation: 150 trials, 28 objects, success within 0.2 m; separate 15-trial obstacle ablation.	Goal success 0.89 versus 0.67 for OWL-v2 + ZoeDepth. With obstacles: arrival 0.60 with versus 0.13 without collision training.	Target arrival and obstacle handling are separate evaluations; visible-target success does not establish open-world search or a collision-risk bound.
TAPM-IVP [22] Table 2	Scene 1 simulation; two robots, ten independent trials per method, common local planner/tracker.	Success 90% for each robot versus 40% and 30% in the two-robot ablation without partitioning and greedy subgoal selection.	The ablation supports the combined design, not either module alone. Rates are per robot; hardware demonstrations use two Ackermann vehicles.
GP-based recovery [23] Fig. 6 and Section V	Ten simulated runs per goal/world; same planner with and without recovery. Counts read from the plotted rates.	Dense 1 m doorways, goal 0: 8/10 versus 3/10 successes. Sparse 2 m doorways, goal 0: 10/10 for both.	Recovery benefit depends on the failure setting. The physical demonstrations do not provide a comparable success-rate estimate.
Home deployment [24] Table 3	Ten-home field deployment; three assistive-service types, including requested but unperformed interventions.	1130 successful, 177 failed, and 232 unperformed interventions out of 1539 requests. Reported success: 86.46% of 1307 attempts.	The success denominator excludes unperformed requests; availability and assistance conditions remain relevant to service-level interpretation.

volume was not established. Overlapping results from different search sources are not independent screening denominators.

S2. ELIGIBILITY AND PUBLICATION SCOPE

Eligible publications include journal articles and main-conference papers concerning wheeled mobile robots or autonomy methods relevant to the navigation problems examined in the survey. Preprints and established conference-to-journal extensions are counted as one work, identified by title, authors, DOI or proceedings identifier, venue, and year. Publication-year statistics use formal volume years, whereas bibliography entries retain publishers' citation years, which can instead refer to online publication.

The selected literature contains 201 journal articles and 44 conference papers: 244 technical studies and one context review on learned navigation [25]. All corpus works are from 2021–2026, with two assessed studies from before 2021 left uncited. The 25 background references outside the corpus comprise representative estimators, related surveys, and foundational sources that attribute the definitions and equations of Section II: the kinematic classification of wheeled robots, the dynamic-window, timed-elastic-band, barrier-function, and Nav2 papers, a SLAM formulation, and a GNSS integrity review.

Of the 244 technical works, 172 were evaluated on target wheeled robots or their data, 12 on road vehicles (mark

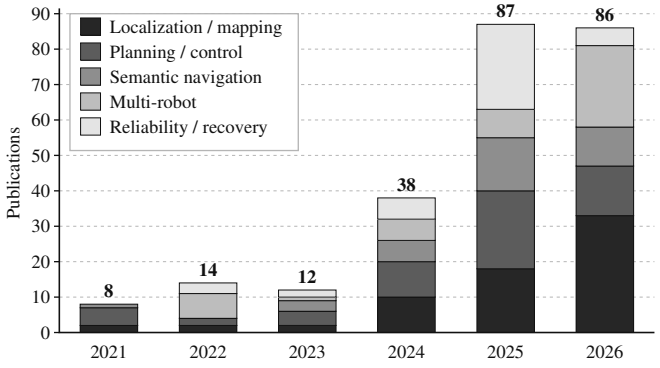


Fig. S1. Selected publications by year and principal navigation function. Each publication appears once. The distribution reflects purposive sampling and topic quotas.

R), 13 on legged or tracked ground robots (G), and 47 on aerial, aquatic, handheld, or mixed carriers or without a specified platform (N). The implementation tables mark the last three groups. Mixed datasets that include a wheeled carrier stay in group N. A sensitivity row adds them to the wheeled subset. The non-wheeled works inform mechanism comparisons when the transfer criteria of Section III hold, with platform differences considered when assessing transfer. The target wheeled works also vary in drivetrain, sensing, and computational resources.

For the technical synthesis, publications are grouped by their principal contribution to localization/mapping, traversability/planning/control, semantic navigation, multi-robot coordination, or fault management. A publication may also inform other areas, although it appears only once in Fig. S1. These five functional groups differ from the six themes used to allocate the thematic sample.

S3. Assessment and Comparison

Comparison attributes include the technical mechanism, platform, sensing, computation, experimental setting, baselines, measured outputs, and limitations. For 23 publications, assessment covered the full paper and numerical tables, whereas for 222 it was limited to selected method, evaluation, and conclusion passages or existing extracts. Of these 222 publications, 87 assessments were based on earlier extractions and 135 on targeted readings. These assessment levels do not provide uniform whole-paper verification across the selected literature. Each value of the evidence coding that is read from a paper carries a source location, either a physical PDF page or a review-card block that, on a light card, cites a page. Values that rest on an earlier extraction were located on the PDF page. Platform group, drivetrain, strongest evidence setting, integrity assessment, and computation placement follow recorded card fields and have no location of their own.

The 48, 36, and 16 works of the last three stages received targeted readings with page-located excerpts matched to both the transcription and PDF text. Their reading records retain mechanism and protocol conditions. Numbers that the survey reports from these works were rechecked in the PDF text, without rendered-table audits. Four citation entries were rewritten from the publisher's citation file, which is kept unchanged, including an early-access year correction and the replacement of an unparsable citation key. One further work has a title that differs between its publisher page and its PDF. The public data record these citation changes separately from the reading depth.

Experimental settings distinguish hardware control, online estimation, recorded robot data, dataset replay, simulation, and numerical studies. Dataset and review publications are identified separately. Tables S3–S7 summarize mechanisms and inputs alongside experimental settings, evaluated quantities, and computation location. Interpretation accounts for external pose, remote computation, low speed, and staged scenes where applicable. Complete-pipeline timing and placement require information beyond processor identity.

Quantitative comparisons retain the reported metrics, baselines, and evaluation conditions. Effect sizes are not pooled across heterogeneous datasets, platforms, or trial units. The evaluation table of the survey counts attributes over the 244 technical publications, excluding the context review. It groups overlapping attributes by measurement level, degradation evidence, same-experiment combination, evidence setting, assessment, and platform group (Section S6). A study may report component and task measures during a degradation test. Purposive selection, uneven assessment depth, and heterogeneous protocols limit the generality of comparative conclusions.

S4. Selected Results and Within-Study Comparisons

Table S2 retains the measures and experimental units of seven selected examples. Numerical results were checked against the cited tables or plots. Each comparison concerns the stated scene, baseline, and information available to the method. The rows therefore do not form a cross-study ranking. The examples include changes in performance ordering, added computation, combined-module ablations, and success denominators to clarify the scope of the conclusions drawn in the survey.

S5. Representative Implementations by Navigation Function

Tables S3–S7 list representative implementations for the five functional sections of the survey. Each row reports the mechanism and its inputs, the experimental setting with the evaluated quantities, and the platform with the computation placement stated by the authors. The rows were chosen to cover the response families discussed in each section, and they are not a ranking or a complete list of the selected works.

S6. Evidence Coding

The codebook, the per-work codes with their source locations, the second coding, the adjudication log, and a list of all 245 works with selection stage, principal section, venue, publisher, and platform group are released as public data with the survey. The coding unit is the publication, whose attributes summarize every experiment it reports. A combined attribute counts only when both parts come from one experiment,

TABLE S3
LOCALIZATION AND MAPPING: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	**Mechanism and inputs**	**Evaluation setting** *Evaluated quantities*	**Platform** *Computation*
Uncertainty-aware localization [26]	Uncertainty-weighted prior-map matching with LIO and gated GNSS. *Inputs:* 3D LiDAR, IMU, GNSS	Acquired data *Pose error and covariance estimates*	wheeled (drive unspecified) *Unspecified*
LODESTAR [18]^{N}	Selective historical anchoring and point selection under degeneracy. *Inputs:* 3D LiDAR, IMU	Replay *Absolute pose error and runtime*	mixed *Desktop evaluation*
ToF-Wheel [27]	Directional process-covariance inflation from geometric corrections. *Inputs:* time-of-flight camera and wheels; LiDAR replaces depth in replay	Acquired data, Replay *Trajectory error across covariance variants*	differential drive *Onboard*
DURAL [28]	Region- and degeneracy-dependent sensor fusion. *Inputs:* LiDAR, IMU, UWB, wheel encoders	Acquired data, Sim. *Localization error across fusion modes*	wheeled (drive unspecified) *Unspecified*
MM-LINS [29]^{R}	Map hibernation, reinitialization, and pose-graph registration. *Inputs:* LiDAR/IMU from handheld and vehicle rigs	Acquired data, Replay *Trajectory error and map registration*	road vehicle *Unspecified*
SE2-LIO [30]	Near-planar pose with stochastic out-of-plane perturbations. *Inputs:* 3D LiDAR, IMU	Online estimate, Acquired data *Pose error; slope-model boundary*	wheeled (drive unspecified) *Onboard*
ProbPer-LiLo [31]^{N}	Semantic-cluster persistence estimation and map filtering. *Inputs:* 3D LiDAR, semantic labels	Acquired data, Replay *Registration and map persistence*	mixed *Unspecified*
DMSA [32]^{N}	Joint grid-distribution fitting over multiple scans. *Inputs:* 3D LiDAR, IMU	Acquired data, Replay *Pose error, runtime, sensor compatibility*	mixed *Desktop evaluation*
Switch-SLAM [33]^{N}	Degeneracy-triggered introduction of visual constraints. *Inputs:* LiDAR, camera, IMU	Replay, Sim. *Trajectory error under modality failure*	mixed *Desktop evaluation*
M2UD [34]	Uneven-terrain localization and mapping benchmark. *Inputs:* LiDAR, RGB-D, IMU, GNSS; mixed reference systems	Dataset, Acquired data *Localization/mapping error and runtime*	wheeled (drive unspecified) *Unspecified*
DR-ICP [35]^{N}	Directional feature selection and weighting. *Inputs:* LiDAR; capture IMU not integrated	Acquired data, Replay, Sim. *Tunnel pose error; pure-LiDAR scope*	mixed *Desktop evaluation*
Decoupled registration [36]^{N}	Subspace decoupling and directional preconditioning. *Inputs:* LiDAR registration constraints	Acquired data, Replay, Numerical *Registration; missing-information boundary*	mixed *Desktop evaluation*
Semantic neural processes [37]^{R}	Semantic neural-process elevation and uncertainty. *Inputs:* LiDAR, image semantics, poses	Acquired data *Elevation error and uncertainty; calibration needed*	road vehicle *Desktop evaluation*

Superscripts mark evaluations on road vehicles (R), other ground robots (G), and non-ground, handheld, mixed, or unspecified carriers (N).

identified by its trial set, table, or figure. Degradation evidence is coded per source in three classes. D1 marks a condition that the authors inject, remove, or vary and report for the same system at a second level, or with and without the evaluated response under the same condition. D2 marks a condition selected for its difficulty without manipulation, including a manipulated condition reported at one level only. D3 marks a condition that occurred during the experiments and is reported without being designed in. A condition mentioned only as motivation is not coded. Moving agents count when the study treats them as a disturbance or varies them. Model error counts only when wrong or out-of-distribution outputs are evaluated.

Nine sources form five groups and violate six assumptions (Fig. 1 of the main paper). A work occupies one matrix cell for every source it addresses, always in the column of its principal section, and counts once in every pair of the functions it codes. Platform groups follow the platform class of the evaluation. The drivetrain follows the class of the target wheeled works. A drivetrain-dependent effect is coded only when a paper reports an effect of slip, turning radius, lateral motion, reversing, or footprint that depends on the drive, which 10 works do. Assessment depth distinguishes full readings, earlier extractions, and targeted readings, as in Section S3.

Tables S8–S10 give the column percentages of the evidence attributes, the works of every matrix cell, and the headline share under alternative definitions and subsets with four ro-

TABLE S4
TRAVERSABILITY, PLANNING, AND CONTROL: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
LRAE [38]	Unknown-region safety classification and route optimization. *Inputs:* 3D LiDAR	Hardware, Sim. *Exploration time and travel cost*	skid steer *Unspecified*
CADNet [39]	Heading-dependent accessible-range prediction for TEB planning. *Inputs:* LiDAR history	Hardware, Replay, Sim. *Traversability prediction and navigation*	wheeled (drive unspecified) *Onboard*
GA-Nav [40]	Semantic terrain segmentation combined with elevation costs. *Inputs:* RGB images, 3D LiDAR	Hardware, Replay *Terrain segmentation and navigation success*	wheeled (drive unspecified) *Onboard*
Anisotropic exploration [41]	Direction-dependent quasi-static terrain costs. *Inputs:* LiDAR terrain map, FAST-LIO pose	Hardware, Sim. *Exploration, travel cost, body roll*	differential drive *Onboard*
FIT-SLAM2 [19]	Keyframe-linked frontiers and information-aware adaptive roadmaps. *Inputs:* RGB-D/LiDAR (simulation); stereo depth (hardware)	Hardware, Sim. *Exploration time, pose error, and runtime*	skid steer *Unspecified*
Vehicle–cloud MPC [42]	Switching between onboard linear and cloud nonlinear MPC. *Inputs:* externally measured poses	Hardware, Sim., Theory *Tracking and collision outcomes*	Ackermann *Mixed placement*
CECBF [43]	CBF constraints with circulation and hold logic. *Inputs:* 2D LiDAR, external robot pose	Hardware *Obstacle avoidance and escape from equilibria*	differential drive *Local offboard*
Safe Horizon MPC [20]	Scenario constraints on joint collision risk with deceleration fallback. *Inputs:* robot/pedestrian poses and velocity estimates	Hardware, Sim. *Task duration and planned collision risk*	skid steer *Unspecified*
Spatiotemporal-Tubes [44]	Adaptive spherical tubes for prescribed-time reach-avoid. *Inputs:* robot state, online unsafe-set geometry	Hardware, Sim., Numerical *Reach-avoid and prescribed-time behavior*	omnidirectional *Unspecified*
Perceive With Confidence [45][G]	Environment-level calibration of obstacle inflation for constrained planning. *Inputs:* RGB-D point clouds, VIO	Hardware, Sim. *Collision-free outcomes and calibration*	legged *Onboard*
Homotopy MPC [46]	Parallel optimization of homotopy alternatives. *Inputs:* externally tracked robot/pedestrian states	Hardware, Sim. *Dynamic avoidance; external pose*	wheeled (drive unspecified) *Unspecified*
Visual-geometric MPPI [47]	Expert-trained cost, speed, and novelty maps for MPPI. *Inputs:* RGB features, LiDAR geometry	Hardware, Acquired data *Interventions, speed, and terrain response*	wheeled (drive unspecified) *Unspecified*
SPARTA [48]	Angle-conditioned risk queried during planning. *Inputs:* prepared point cloud, LiDAR/IMU pose	Hardware, Sim. *Simulated tire risk; virtual risky hardware branch*	skid steer *Onboard*

Superscripts mark evaluations on road vehicles (R), other ground robots (G), and non-ground, handheld, mixed, or unspecified carriers (N).

bustness findings fixed before primary coding. Findings RF1 (localization has the lowest headline share) and RF4 (task outcomes in at most 5% of localization works) hold under every definition and in every stage, depth, and quota subset, with a tie among full readings. RF1 fails only among works from 2026. RF4 fails only in one quota theme. RF2 (fault management has the highest share) and RF3 (task outcomes in at least 90% of planning and semantic works) fail under the final codes. Planning has the highest share. Task outcomes appear in 44 of 56 planning and 33 of 36 semantic works. The card-level coding of the earlier draft had supported RF2.

S7. LANGUAGE-MODEL SECOND CODING

A stratified random sample of 40 technical works, eight per principal section, was drawn with a seed fixed in the codebook before primary coding. Within each section, the draw takes two full readings and splits the other six between earlier extractions and targeted readings by their section shares. The draw was repeated until every selection stage appeared. A separate language-model session, configured for a different model, coded the thirteen double-coded attributes from the codebook, the sample list, and the review cards and transcriptions of the sampled works. It had no access to the primary codes, which another session wrote for all 245 works and froze by hash. The public coding record names the model configured for each pass and gives the timeline of the two passes. Neither session recorded the model that actually ran.

Table S11 reports percent agreement and unweighted Cohen's kappa before adjudication. Both coders are language-model sessions that read review cards themselves prepared with language-model tools. The statistics therefore describe how consistently two such coders apply the codebook to shared evidence and are not human inter-rater reliability. Among the attributes, kappa is lowest for the drivetrain-dependent

TABLE S5
SEMANTIC AND LANGUAGE-GUIDED NAVIGATION: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	**Mechanism and inputs**	**Evaluation setting** *Evaluated quantities*	**Platform** *Computation*
VLM-Social-Nav [49]	Cue-triggered VLM judgments used as social DWA costs. *Inputs:* RGB images, LiDAR	Hardware *Task success, collisions, and social behavior*	differential drive *Mixed placement*
Co-NavGPT [50][G]	VLM frontier allocation with geometric goal execution. *Inputs:* RGB-D, shared map, robot states	Hardware, Sim. *HM3D search; separate hardware demos*	legged *Onboard pose; offboard perception/planning*
OpenIN [51]	Carrier–object memory with visibility-dependent updates. *Inputs:* RGB-D, language/image queries, SLAM pose	Hardware, Sim. *Success across sequential object queries*	wheeled (drive unspecified) *Unspecified*
VLFM [52][G]	Image–language similarity projected into a frontier value map. *Inputs:* RGB-D, odometry	Hardware, Sim. *Goal success and path efficiency*	legged *Unspecified*
HOV-SG [53][G]	Hierarchical retrieval over floor, room, and object graphs. *Inputs:* RGB-D, LiDAR-derived poses	Hardware, Sim., Replay *Object retrieval and navigation success*	legged *Unspecified*
Uni-NaVid [54][G]	Multiscale video memory and discrete action prediction. *Inputs:* monocular RGB video, instruction	Hardware, Sim. *Goal success and path efficiency*	legged *Local offboard*
LeLaN [21]	Receding-horizon language-conditioned velocity prediction. *Inputs:* RGB image, text instruction	Hardware, Replay *Target arrival and action-generation time*	differential drive *Onboard*
VL-TGS [55]	VLM selection among generated trajectories with local DWA control. *Inputs:* LiDAR, RGB, velocity, GPS	Hardware, Acquired data *Outdoor traversal and model-response time*	wheeled (drive unspecified) *Mixed placement*
GSON [56]	Group-aware routing with predictive barrier-constrained control. *Inputs:* 2D LiDAR, RGB images	Hardware, Replay *Social disturbance, travel time, and jerk*	differential drive *Unspecified*
AVLMaps [17]	Multimodal spatial retrieval and goal APIs. *Inputs:* offline RGB-D/pose/audio map, instructions	Hardware, Sim. *Multimodal goals; prepared-map validity*	wheeled (drive unspecified) *Unspecified*
VLM-GroNav [57]	Interaction-grounded terrain costs and DWA. *Inputs:* RGB, LiDAR, proprioception, GPS, aerial images	Hardware *Outdoor task outcomes; GPS/cloud dependencies*	skid steer *Mixed placement*
SocialNav-SUB [58]	Social scene-understanding VQA benchmark. *Inputs:* recorded images, pedestrian annotations	Dataset *Human/rule agreement; no closed-loop control*	wheeled (drive unspecified) *Mixed placement*

Superscripts mark evaluations on road vehicles (R), other ground robots (G), and non-ground, handheld, mixed, or unspecified carriers (N).

effect, which the second coder assigned to four works and the primary coder to one, and for the nominal control. The maximum of 1.00 belongs to incidental sources and rests on the same 3 of 360 source flags set by both coders. A third language-model session with access to both codings and the sources re-read every one of the 112 differing values at both locators and logged a reason and the deciding location. It kept the primary value 58 times, adopted the second value 46 times, and revised both 8 times. Codebook version 1.1 clarified the two rules that had caused disagreement in at least three works, by limiting the response family to its primary value and by narrowing the drivetrain-dependent effect. That effect now excludes differences between wheeled and other locomotion, limits that apply to every wheel arrangement, and kinematic models without a reported geometry-dependent result. A recheck of these rules across the corpus changed six further values. After an internal re-review, a further recheck searched all works for drivetrain-dependent effects that the primary codes might have missed, through a keyword screen of every review card and transcription and a reading of each candidate passage on its PDF page. It added 4, for 10 works in all. The same recheck read the PDF pages of the 19 works coded at the physical evidence level without a physical task outcome under a named condition. It kept the level where the physical run states a result under that condition, lowered it for 2 works, and corrected the source locations of 2 works. For 5 of the kept works, the text or a table of the physical run reports a task outcome under that condition. The recheck coded these combinations as physical. A consistency check of the final codes now requires a stated result for the physical level. Adjudication changed the headline count from 81 to 81.

S8. DERIVATION OF THE OPEN INTERFACES

A candidate interface is an unordered pair of distinct functions with one coupling source. Its works code both functions and address, name, or report the source. The rule, fixed in codebook version 1.0 before coding (commit 67872f4), keeps candidates with at least three works and sorts them by the number of works without direct joint wheeled evidence, then by safety relevance (planning or fault management in the pair),

TABLE S6
MULTI-ROBOT COORDINATION: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
ROAM [59]	One-hop consensus for semantic mapping and viewpoint planning. *Inputs:* RGB-D, LiDAR	Hardware, Sim. *Distributed mapping and exploration*	skid steer *Onboard*
Active graph exploration [60]^{N}	Submodular loop-edge selection after coverage planning. *Inputs:* prior topo-metric graph	Sim. *Simulated coverage and graph objective*	none *Desktop evaluation*
Range integrity [61]^{N}	Range-consistency localization-error reconstruction. *Inputs:* emulated ranges, reported poses	Online monitor (mixed reality), Sim. *Error reconstruction with emulated ranges*	aerial *Central ground station; emulated ranges*
Commerge [62]	Descriptor matching and scan selection before submap registration. *Inputs:* 3D LiDAR, LIO poses	Acquired data, Replay *Map-merging error and transmitted data*	wheeled (drive unspecified) *Desktop evaluation*
iMESA [63]^{N}	Incremental local factor graphs with shared-variable consensus. *Inputs:* odometry and inter-robot factors	Replay, Numerical *Estimation error with limited communication*	mixed *Desktop evaluation*
Collaborative map repair [64]^{R}	Residual-based fault isolation and correlation-aware map repair. *Inputs:* LiDAR, wheel odometry; GNSS initialization	Acquired data, Sim. *Pose/map error and fault isolation*	road vehicle *Unspecified*
TAPM-IVP [22]	Terrain-aware 3D Voronoi allocation and subgoal selection. *Inputs:* LiDAR, A-LOAM poses	Hardware, Sim. *Success, travel cost, and path duplication*	Ackermann *Onboard*
Failure-aware tracking [65]^{N}	Failure-dependent tracking modes and target-state fusion. *Inputs:* target range/bearing and inter-robot messages	Hardware, Sim. *Target-state error during failures*	aerial *Unspecified*
Cross-rank [66]^{G}	Intersection allocation from shared position histories. *Inputs:* LiDAR localization and low-bandwidth radio	Hardware, Sim. *Exploration under intermittent reception*	legged *Mixed placement*
CoVOR-SLAM [67]	Similarity-pose-graph fusion for monocular scale correction. *Inputs:* camera, inter-rover UWB ranges	Acquired data, Replay, Sim. *Pose error and monocular scale correction*	wheeled (drive unspecified) *Desktop evaluation*
Swarm-SLAM [68]	Sparse loop prioritization and decentralized inference. *Inputs:* local odometry, loop observations	Online estimate, Replay *Online collaborative estimation; team exchange*	skid steer *Onboard*
Planned synchronization [69]	Joint motion and observation scheduling. *Inputs:* relative marker observations, IMU	Hardware, Numerical *Formation and check-in cost; modeled non-idealities*	differential drive *Onboard*
Bandwidth-efficient SLAM [70]	Lossy keyframe selection and lossless coding. *Inputs:* visual/inertial keyframes	Online estimate, Replay *Bandwidth/accuracy tradeoff; central back end*	wheeled (drive unspecified) *Mixed placement*

Superscripts mark evaluations on road vehicles (R), other ground robots (G), and non-ground, handheld, mixed, or unspecified carriers (N).

then by the number of target wheeled works. At equal direct evidence, this count ranks a candidate with more works above one with fewer, with candidate sizes shaped by the selection stages, including the integration-questions stage acquired for the interfaces. Table S12 gives the function-pair co-occurrence with evidence levels. Table S13 ranks the candidates at or above the lowest developed interface, with every interface cell, two sensitivity ranks, and the reason for each candidate that is not developed.

Each interface takes the rank of its best eligible cell. Interface 4 has rank 1 (LOC-MRC:COM), Interface 1 rank 4 (LOC-REL:PER), Interface 2 rank 10 (PLN-SEM:MOD), and Interface 3 rank 11 (LOC-PLN:PER) among 32 eligible candidates. The computation-limit cell of Interface 2 (PLN-SEM:RES) ranks 26, whereas the terrain cell of Interface 3 (LOC-PLN:TER) has 2 works, below the eligibility threshold. The four interfaces therefore do not form the top of the ranking. The main paper presents them as an agenda proposed by the survey. Five of the 6 undeveloped candidates above Interface 3 are merged into Interfaces 4 or 2, whose comparisons vary their sources. The sixth, actuator faults within one tracking controller, is named as a further gap. The alternatives named by the internal review either rank below Interface 3 (map change forcing plan re-validation, perception latency, and energy or computation limits on planning) or fall outside the function pairs (human assistance, coded in 10 works). Removing the integration-questions stage, which was acquired for the interfaces, gives ranks 4, 8, 9, and 1 for Interfaces 1–4. Sorted by gap share (Rank B), Interfaces 1–4 rank 5, 26, 18, and 12, below the undeveloped actuator-fault cell PLN-REL:ACT. The public data list the contributing works of every interface with platform group and evidence level. Interfaces 1–

TABLE S7
RELIABILITY AND RECOVERY: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
Visual NMPC recovery [71]	Tracking-loss stop and relocalization; restart after failure. *Inputs:* stereo vision, actuator feedback	Hardware *Tracking; stop/relocalization handling*	skid steer *Unspecified*
Set-based LPV diagnosis [72]^{R}	Zonotopic detection and fault-class separation. *Inputs:* yaw rate, odometry, sideslip, steering angle	Acquired data *Detectability and isolability under injected faults*	road vehicle *Desktop evaluation*
Reliable MCL [73]	Joint pose/reliability estimation with global relocalization. *Inputs:* 2D LiDAR, odometry or inertial navigation	Acquired data, Replay, Sim. *Reliability estimates and relocalization*	wheeled (drive unspecified) *Unspecified*
Long-term home deployment [24]	Long-term service, docking, and remote human supervision. *Inputs:* LiDAR, RGB-D, docking and home sensors	Hardware *Service outcomes and interruptions*	wheeled (drive unspecified) *Mixed placement*
KSFD [74]	Kinematic residuals for sensor-fault diagnosis. *Inputs:* inertial channels and an inertial-aided position substitute	Acquired data, Numerical *Fault-isolation residual patterns*	Ackermann *Desktop evaluation*
Behavior-tree recovery [75]^{N}	Behavior-tree supervision and failure-specific recovery. *Inputs:* terrain maps, contacts, planner status	Hardware, Sim. *Recovery on retained scenarios; human-rescue boundary*	mixed *Onboard*
Line-map integrity [76]	Prior-line-map localization with an error protection level. *Inputs:* monocular images, IMU, prior 3D lines	Acquired data, Replay, Sim. *Pose error and protection-level coverage*	wheeled (drive unspecified) *Desktop evaluation*
Event-triggered tracking [77]	Observer-based fault estimation and event-triggered tracking. *Inputs:* vehicle-posture outputs in the model	Hardware-in-the-loop, Numerical *Tracking error and update frequency*	Ackermann *Mixed placement*
Sentinel [78]	Action-consistency and VLM progress monitoring. *Inputs:* external RGB-D and mobile-base motion capture	Online estimate, Sim. *Failure detection and nominal alarms*	omnidirectional *Mixed placement*
RTEB [79]	Hybrid A* reinitialization and local smoothing. *Inputs:* RTK pose, voxel obstacle map	Hardware, Sim. *Planner recovery; TEB/NMPC comparisons*	wheeled (drive unspecified) *Onboard*
SCALE [80]	Affordance prediction and anti-novelty recovery. *Inputs:* RGB at deployment; offline robot data	Hardware *Visual recovery and intervention need*	skid steer *Onboard*
LiReN [81]	Online RL with resets, goals, and charging. *Inputs:* RGB policy; LiDAR/map, bumper for supervision	Hardware *Adaptation with operational resets and infrastructure*	differential drive *Mixed placement*

Superscripts mark evaluations on road vehicles (R), other ground robots (G), and non-ground, handheld, mixed, or unspecified carriers (N).

4 have 1, 6, 1, and 0 works with direct joint wheeled evidence.

REFERENCES

[1] D. Shah, B. Osiński, B. Ichter, and S. Levine, "LM-Nav: Robotic navigation with large pre-trained models of language, vision, and action," in *Proceedings of The 6th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, K. Liu, D. Kulic, and J. Ichnowski, Eds., vol. 205. PMLR, 14–18 Dec 2023, pp. 492–504. [Online]. Available: https://proceedings.mlr.press/v205/shah23b.html

[2] D. Shah, A. Sridhar, A. Bhorkar, N. Hirose, and S. Levine, "GNM: A general navigation model to drive any robot," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 7226–7233.

[3] D. Shah *et al.*, "ViNT: A foundation model for visual navigation," in *Proceedings of The 7th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, J. Tan, M. Toussaint, and K. Darvish, Eds., vol. 229. PMLR, 06–09 Nov 2023, pp. 711–733. [Online]. Available: https://proceedings.mlr.press/v229/shah23a.html

[4] A. Sridhar, D. Shah, C. Glossop, and S. Levine, "NoMaD: Goal masked diffusion policies for navigation and exploration," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 63–70.

[5] Q. Gu *et al.*, "ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 5021–5028.

[6] D. D. Fan, K. Otsu, Y. Kubo, A. Dixit, J. Burdick, and A. akbar Agha-mohammadi, "STEP: Stochastic traversability evaluation and planning for risk-aware off-road navigation," in *Proceedings of Robotics: Science and Systems*, Virtual, July 2021.

[7] X. Cai *et al.*, "EVORA: Deep evidential traversability learning for risk-aware off-road autonomy," *IEEE Transactions on Robotics*, vol. 40, pp. 3756–3777, 2024.

[8] Y. Tian, Y. Chang, F. Herrera Arias, C. Nieto-Granda, J. P. How, and L. Carlone, "Kimera-Multi: Robust, distributed, dense metric-semantic SLAM for multi-robot systems," *IEEE Transactions on Robotics*, vol. 38, no. 4, pp. 2022–2038, 2022.

[9] Y. Chang *et al.*, "LAMP 2.0: A robust multi-robot SLAM system for operation in challenging large-scale underground environments," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 9175–9182, 2022.

[10] N. Hudson *et al.*, "Heterogeneous ground and air platforms, homogeneous sensing: Team CSIRO Data61's approach to the DARPA Subterranean Challenge," *Field Robotics*, vol. 2, pp. 595–636, 2022.

[11] S. Scherer *et al.*, "Resilient and modular subterranean exploration with a team of roving and flying robots," *Field Robotics*, vol. 2, pp. 678–734, 2022.

[12] M. Tranzatto *et al.*, "CERBERUS in the DARPA Subterranean

TABLE S8
EVIDENCE ATTRIBUTES OF THE TECHNICAL WORKS BY PRINCIPAL SECTION. EACH PERCENTAGE REFERS TO THE WORKS OF ITS COLUMN. ATTRIBUTES OVERLAP.

Attribute	IV	V	VI	VII	VIII	All
Publications	67	56	36	45	40	244
Task outcome reported	1 (1%)	44 (79%)	33 (92%)	25 (56%)	20 (50%)	123 (50%)
Task outcome in simulation only	0 (0%)	19 (34%)	9 (25%)	10 (22%)	11 (28%)	49 (20%)
Physical task outcome	1 (1%)	25 (45%)	24 (67%)	15 (33%)	9 (22%)	74 (30%)
Degradation source addressed	65 (97%)	53 (95%)	24 (67%)	38 (84%)	39 (98%)	219 (90%)
Named condition (D1 or D2)	61 (91%)	46 (82%)	19 (53%)	27 (60%)	34 (85%)	187 (77%)
Injected or varied with a comparator (D1)	16 (24%)	15 (27%)	6 (17%)	13 (29%)	11 (28%)	61 (25%)
Selected condition only (D2)	45 (67%)	31 (55%)	13 (36%)	14 (31%)	23 (58%)	126 (52%)
Incidental condition only (D3)	1 (1%)	1 (2%)	2 (6%)	4 (9%)	2 (5%)	10 (4%)
Named condition with a task outcome in the same experiment	1 (1%)	36 (64%)	17 (47%)	11 (24%)	16 (40%)	81 (33%)
D1 condition with a task outcome in the same experiment	0 (0%)	12 (21%)	6 (17%)	5 (11%)	4 (10%)	27 (11%)
Same-experiment combination on a physical robot	1 (1%)	20 (36%)	13 (36%)	2 (4%)	7 (18%)	43 (18%)
Direct joint wheeled evidence	1 (1%)	17 (30%)	12 (33%)	1 (2%)	6 (15%)	37 (15%)
Named condition, physical closed loop	1 (1%)	24 (43%)	14 (39%)	5 (11%)	11 (28%)	55 (23%)
Named condition, online hardware or recorded data	46 (69%)	3 (5%)	0 (0%)	6 (13%)	11 (28%)	66 (27%)
Named condition, replay, simulation, numerical model or resource	14 (21%)	19 (34%)	5 (14%)	16 (36%)	12 (30%)	66 (27%)
Integrity or failure-risk assessment	3 (4%)	1 (2%)	0 (0%)	1 (2%)	9 (22%)	14 (6%)
Computation placement reported	45 (67%)	31 (55%)	18 (50%)	33 (73%)	28 (70%)	155 (64%)
Target wheeled robot	30 (45%)	50 (89%)	30 (83%)	32 (71%)	30 (75%)	172 (70%)
Road vehicle	7 (10%)	1 (2%)	0 (0%)	0 (0%)	4 (10%)	12 (5%)
Other ground robot	1 (1%)	4 (7%)	5 (14%)	3 (7%)	0 (0%)	13 (5%)
Non-ground, handheld, mixed-carrier or no platform	29 (43%)	1 (2%)	1 (3%)	10 (22%)	6 (15%)	47 (19%)
Drivetrain-dependent effect reported	3 (4%)	3 (5%)	0 (0%)	0 (0%)	4 (10%)	10 (4%)

TABLE S9
WORKS THAT ADDRESS EACH DEGRADATION SOURCE, BY PRINCIPAL SECTION. A WORK APPEARS IN THE ROW OF EVERY SOURCE IT ADDRESSES AND ALWAYS IN THE COLUMN OF ITS PRINCIPAL SECTION.

Source (assumption)	IV	V	VI	VII	VIII
Perceptual degradation (A1)	[13], [18], [26]–[29], [33]–[36], [82]–[109]	[19], [110], [111]	[112], [113]	[12], [114]–[116]	[71], [117]–[121]
Sensor fault and false association (A1)	[64], [92], [122]–[125]	[126]	[2]	[8], [9], [61], [65], [68], [114], [127], [128]	[72], [74], [76], [77], [129]–[134]
Terrain and wheel-ground contact (A2)	[27], [28], [30], [34], [37], [84], [99], [101], [102], [105], [107], [122], [135]–[139]	[6], [7], [14], [15], [19], [38]–[41], [47], [48], [140]–[148]	[55], [57], [112], [149]	[12], [22], [150]	[71], [75], [134], [151], [152]
Moving agents (A3)	[26], [32], [34], [86], [91], [97], [108], [137], [153]–[160]	[20], [39], [42]–[44], [46], [111], [141], [161]–[175]	[21], [49], [56], [58], [176]–[179]	–	[24], [73], [79], [152], [180]–[183]
Map and scene change (A3)	[26], [31], [64], [89], [95], [98], [108], [124], [125], [137], [155], [158], [160], [184]–[186]	[172], [187]	[5], [51], [113], [177], [188]–[190]	[62]	[24], [73], [81], [191]
Learned and semantic inference error (A4)	[37], [99], [107], [192]	[7], [45], [47], [140], [162], [166], [170], [173]–[175], [193]–[196]	[2], [3], [58], [149], [179], [197], [198]	–	[78], [80], [81], [119], [152], [180], [182], [183], [199]
Communication loss, delay and bandwidth (A5)	[64], [67]	[42], [126], [171]	–	[8]–[12], [59], [62], [63], [65], [66], [68]–[70], [116], [128], [150], [200]–[213]	[77], [214]
Actuator fault and saturation (A6)	–	[168], [215]–[218]	[2]	[69], [200], [219]–[221]	[72], [77], [132], [133], [222]–[225]
Computation and energy limits (A6)	[106], [137], [156], [186]	[42], [126], [175], [193]	[5], [21], [53]–[56], [113], [179], [188]	[62], [150], [201], [203], [206], [208]	[81], [117], [120], [214], [226]–[228]

Challenge," *Science Robotics*, vol. 7, no. 66, p. eabp9742, 2022. [Online]. Available: https://www.science.org/doi/abs/10.1126/scirobotics.abp9742

[13] S. Zhao *et al.*, "SubT-MRS dataset: Pushing SLAM towards all-weather environments," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2024, pp. 22 647–22 657.

[14] P. Jiang, P. Osteen, M. Wigness, and S. Saripalli, "RELLIS-3D dataset: Data, benchmarks and analysis," in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 1110–1116.

TABLE S10
SENSITIVITY OF THE SHARE OF TECHNICAL WORKS THAT REPORT A NAMED DEGRADATION CONDITION AND A TASK OUTCOME IN THE SAME EXPERIMENT. ROWS VARY THE DEFINITION, THE CODING UNIT OR THE SUBSET. RF1-RF4 STATE WHETHER EACH FINDING FIXED IN CODEBOOK VERSION 1.0 BEFORE CODING HOLDS.

Row	Works	%	IV %	V %	VI %	VII %	VIII %	RF1	RF2	RF3	RF4
S-def-headline	81/244	33	1	64	47	24	40	yes	no	no	yes
S-def-D1	27/244	11	0	21	17	11	10	yes	no	no	yes
S-def-D1-nominal	11/244	5	0	5	11	7	2	yes	no	no	yes
S-def-physical	43/244	18	1	36	36	4	18	yes	no	no	yes
S-def-with-incidental	88/244	36	1	66	53	31	42	yes	no	no	yes
S-unit-any	83/244	34	1	66	47	24	42	yes	no	no	yes
S-card-round3	61/244	25	3	36	25	29	42	yes	yes	no	yes
S-depth-full	7/23	30	0	80	67	0	20	tie	no	no	yes
S-depth-light-legacy	33/86	38	0	62	58	40	22	yes	no	no	yes
S-depth-light-targeted	41/135	30	3	64	38	19	50	yes	no	no	yes
S-depth-no-legacy	48/158	30	2	67	42	17	45	yes	no	no	yes
Without Initial sample	63/199	32	2	61	42	28	47	yes	no	no	yes
Without Complementary publications	79/240	33	1	64	47	23	41	yes	no	no	yes
Without Thematic sample	61/184	33	2	67	46	18	39	yes	no	no	yes
Without Integration questions	76/229	33	2	64	47	20	43	yes	no	no	yes
Without Targeted IEEE coverage	75/224	33	2	63	49	26	41	yes	no	no	yes
Without Further targeted coverage	68/196	35	2	67	50	31	29	yes	no	no	yes
Without Recent IEEE coverage	68/208	33	0	63	47	24	38	yes	no	no	yes
Without Coverage additions	77/228	34	2	65	50	28	40	yes	no	no	yes
S-year-2021-2023	10/33	30	0	60	50	0	40	tie	no	no	yes
S-year-2024	11/38	29	0	80	17	17	17	yes	no	no	yes
S-year-2025	40/87	46	0	64	60	50	54	yes	no	no	yes
S-year-2026	20/86	23	3	57	45	26	0	no	no	no	yes
S-quota	33/96	34	3	61	50	39	46	yes	no	no	yes
S-non-quota	48/148	32	0	66	45	15	37	yes	no	no	yes
S-quota-theme-T1	0/16	0	0	–	–	–	–	n/a	n/a	n/a	yes
S-quota-theme-T2	2/16	12	8	33	–	–	–	n/a	n/a	n/a	no
S-quota-theme-T3	10/16	62	–	67	–	0	–	n/a	n/a	n/a	n/a
S-quota-theme-T4	6/16	38	–	–	–	38	–	n/a	n/a	n/a	n/a
S-quota-theme-T5	8/16	50	–	–	50	–	–	n/a	n/a	n/a	n/a
S-quota-theme-T6	7/16	44	0	–	–	100	46	n/a	n/a	n/a	n/a
S-no-ieee-only	62/188	33	0	62	48	25	38	yes	no	no	yes
S-no-coverage	77/228	34	2	65	50	28	40	yes	no	no	yes
S-platform-WHL	69/172	40	3	64	53	25	40	yes	no	no	yes
S-platform-WHL-mixed	69/182	38	3	64	53	24	40	yes	no	no	yes
S-platform-non-WHL	12/72	17	0	67	17	23	40	yes	no	no	yes

Definition rows change the measure and use all technical works. Other rows apply the headline measure to a subset. A finding is n/a when a function it compares has fewer than three works in the subset. All shares describe the purposively selected sample.

[15] S. Triest, M. Sivaprakasam, S. J. Wang, W. Wang, A. M. Johnson, and S. Scherer, "TartanDrive: A large-scale dataset for learning off-road dynamics models," in *2022 International Conference on Robotics and Automation (ICRA)*, 2022, pp. 2546–2552.

[16] S. K. Ramakrishnan *et al.*, "Habitat-Matterport 3D dataset (HM3D): 1000 large-scale 3D environments for embodied AI," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, J. Vanschoren and S. Yeung, Eds., vol. 1, 2021. [Online]. Available: https://datasets-benchmarks-proceedings.neurips.cc/paper_files/paper/2021/file/34173cb38f07f89ddbebc2ac9128303f-Paper-round2.pdf

[17] C. Huang, O. Mees, A. Zeng, and W. Burgard, "Multimodal spatial language maps for robot navigation and manipulation," *The International Journal of Robotics Research*, vol. 0, no. 0, p. 02783649251351658, 2025. [Online]. Available: https://doi.org/10.1177/02783649251351658

[18] E. M. Lee, K. C. Marsim, and H. Myung, "LODESTAR: Degeneracy-aware LiDAR-inertial odometry with adaptive Schmidt-Kalman filter and data exploitation," *IEEE Robotics and Automation Letters*, vol. 11, no. 1, pp. 922–929, 2026.

[19] S. Saravanan, A. Bains, C. P.C. Chanel, and D. Vivet, "FIT-SLAM 2: Efficient 3D exploration with Fisher information and traversability-based adaptive roadmap," *Robotics and Autonomous Systems*, vol. 194, p. 105188, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0921889025002854

[20] O. de Groot, L. Ferranti, D. M. Gavrila, and J. Alonso-Mora, "Scenario-based motion planning with bounded probability of collision," *The International Journal of Robotics Research*, vol. 44, no. 9, pp. 1507–1525, 2025. [Online]. Available: https://doi.org/10.1177/02783649251315203

[21] N. Hirose, C. Glossop, A. Sridhar, O. Mees, and S. Levine, "LeLaN: Learning a language-conditioned navigation policy from in-the-wild video," in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, P. Agrawal, O. Kroemer, and W. Burgard, Eds., vol. 270. PMLR, 06–09 Nov 2025, pp. 666–688. [Online]. Available: https://proceedings.mlr.press/v270/hirose25b.html

[22] H. Zhao, K. Xu, N. Xu, X. Li, and J. Jin, "Multi-robot collaborative navigation framework based on 3D Voronoi partitioning in uneven and unstructured environments," *Journal of Field Robotics*, vol. 43, no. 7, pp. 4531–4548, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.70260

[23] N. Mohammad, J. Higgins, and N. Bezzo, "A GP-based robust motion planning framework for agile autonomous robot navigation and recovery in unknown environments," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 2418–2424.

[24] M. Luperto *et al.*, "User feedback and remote supervision for assisted living with mobile robots: A field study in long-term autonomy," *Robotics and Autonomous Systems*, vol. 155, p. 104170, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0921889022000963

TABLE S11
AGREEMENT BETWEEN THE INDEPENDENT SECOND CODING AND THE FROZEN PRIMARY CODES ON THE 40 SAMPLED WORKS, BEFORE ADJUDICATION. BOTH CODERS ARE LANGUAGE-MODEL SESSIONS CONFIGURED FOR DIFFERENT MODELS. THE VALUES THEREFORE DESCRIBE HOW CONSISTENTLY TWO SUCH CODERS APPLY THE CODEBOOK TO SHARED EVIDENCE. THEY ARE NOT HUMAN INTER-RATER RELIABILITY.

Attribute	Unit	n	Cat.	Agr. %	κ	Exact %	Flag κ	Works	P/S/B
Functions	non-principal function flags, pooled	160	2	96.2	0.89	85.0	0.75–1.00	6	2/4/0
Addressed sources	source flags, pooled	360	2	96.4	0.86	70.0	0.66–1.00	12	4/8/0
Manipulated sources (D1)	source flags, pooled	360	2	94.7	0.53	67.5	–	13	4/8/1
Selected sources (D2)	source flags, pooled	360	2	95.8	0.61	70.0	–	12	4/7/1
Incidental sources (D3)	source flags, pooled	360	2	100.0	1.00	100.0	–	0	0/0/0
Named sources (D1 or D2)	source flags, pooled	360	2	97.2	0.86	80.0	0.63–1.00	8	–
Response family	primary family	40	20	85.0	0.84	–	–	6	–
Response family	full value, with the optional secondary family of version 1.0	40	46	30.0	0.29	–	–	28	24/0/4
Publication class	strongest class	40	3	72.5	0.58	–	–	11	–
Nominal control	value	40	3	67.5	0.42	–	–	13	4/9/0
Single-level injection	value	40	2	92.5	0.54	–	–	3	2/1/0
Task setting	value	40	4	90.0	0.86	–	–	4	1/2/1
Same-experiment class	value	40	3	82.5	0.69	–	–	7	4/3/0
Same-experiment setting	value	40	3	92.5	0.86	–	–	3	1/2/0
Degradation evidence level	value	40	7	85.0	0.81	–	–	6	3/2/1
Drivetrain effect	value	40	2	87.5	−0.04	–	–	5	5/0/0

n counts the scored units and Cat. the categories that either coder used. κ is unweighted Cohen's kappa (n/a when both coders use one category). Set-valued attributes are scored as one binary flag per value. For them, κ and Agr. pool all flag decisions of the attribute (for functions the four non-principal functions of each work). Exact gives the share of works with identical sets. Flag κ gives the range of the per-flag kappas (functions over the 32 works whose principal function differs, sources over all 40 works). The unit of response family fixed in codebook version 1.0 is the primary family. The second row compares the full value, which in codebook version 1.0 could add an optional secondary family. Named sources and publication class are derived from the three class columns. Works counts the works with differing values and P/S/B how the logged disagreements were resolved: primary kept, second adopted, both revised.

TABLE S12
FUNCTION-PAIR CO-OCCURRENCE WITH EVIDENCE LEVELS. NAMED, D1 AND PHYSICAL COUNT WORKS WITH A SAME-EXPERIMENT COMBINATION OF A NAMED CONDITION AND A TASK OUTCOME, A D1 CONDITION, OR A PHYSICAL ROBOT. WHEELED JOINT COUNTS SUCH PHYSICAL COMBINATIONS ON TARGET WHEELED ROBOTS.

Pair	Works	Named	D1	Physical	Wheeled joint	Wheeled	Level
LOC–PLN	14	5	3	1	1	12	P
LOC–SEM	8	2	2	2	2	5	P
LOC–MRC	25	1	1	0	0	18	P
LOC–REL	19	2	0	1	1	11	P
PLN–SEM	20	14	3	10	9	18	P
PLN–MRC	23	12	6	3	2	20	P
PLN–REL	25	15	2	9	7	19	P
SEM–MRC	1	0	0	0	0	0	none
SEM–REL	1	1	0	0	0	0	R
MRC–REL	12	5	5	0	0	6	P

[25] K. Zhu and T. Zhang, "Deep reinforcement learning based mobile robot navigation: A review," *Tsinghua Science and Technology*, vol. 26, no. 5, pp. 674–691, 2021.

[26] G. Park and W. Chung, "Uncertainty-aware LiDAR-based localization for outdoor mobile robots," *Journal of Field Robotics*, vol. 41, no. 8, pp. 2790–2804, 2024. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.22392

[27] X. Yang, W. Li, Z. Wei, Z. Li, L. Huang, and B. Wang, "ToF-wheel odometry: Navigating degeneration of localization with limited sensing capabilities," *IEEE Robotics and Automation Letters*, vol. 11, no. 10, pp. 11 410–11 417, 2026.

[28] K. Hu, M. Li, Z. Jin, C. Tang, E. Hu, and G. Zhou, "DURAL: Degradation-resistant robust adaptive localization by LiDAR-inertial-UWB-wheel fusion for coal mine robots," *Journal of Field Robotics*, vol. 43, no. 5, pp. 3559–3577, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.70200

[29] Y. Ma *et al.*, "MM-LINS: A multi-map LiDAR-inertial system for over-degenerate environments," *IEEE Transactions on Intelligent Vehicles*, vol. 10, no. 1, pp. 472–482, 2025.

[30] J. Chen, H. Wang, M. Hu, and P. N. Suganthan, "Versatile LiDAR-inertial odometry with SE(2) constraints for ground vehicles," *IEEE Robotics and Automation Letters*, vol. 8, no. 6, pp. 3486–3493, 2023.

[31] W. Ali, Y. Cai, P. Jensfelt, and T.-M. Nguyen, "ProbPer-LiLo: Probabilistic persistency modeling for life-long mapping," *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 2530–2537, 2026.

[32] D. Skuddis and N. Haala, "DMSA - dense multi scan adjustment for LiDAR inertial odometry and global optimization," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 12 027–12 033.

[33] J. Lee *et al.*, "Switch-SLAM: Switching-based LiDAR-inertial-visual SLAM for degenerate environments," *IEEE Robotics and Automation Letters*, vol. 9, no. 8, pp. 7270–7277, 2024.

[34] Y. Jia, S. Wang, S. Shao, Y. Wang, F. Zhang, and T. Wang, "M2UD: A multi-model, multi-scenario, uneven-terrain dataset for ground robot with localization and mapping evaluation," *The International Journal of Robotics Research*, vol. 45, no. 10, pp. 1457–1477, 2026. [Online]. Available: https://doi.org/10.1177/02783649251401224

[35] G. Qian *et al.*, "DR-ICP: Robust LiDAR SLAM registration for degenerate tunnels," *Journal of Field Robotics*, oct 2026, early access; published online 12 October 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/rob.70362

[36] X. Hu, X. Chen, M. Jia, J. Wu, P. Tan, and S. L. Waslander, "DCReg: Decoupled characterization for efficient degenerate LiDAR registration," *The International Journal of Robotics Research*, vol. 0, no. 0, p. 02783649261465040, 2026. [Online]. Available: https://doi.org/10.1177/02783649261465040

[37] S. Jung, D. Gwak, B. Boots, and J. Hays, "Uncertainty-aware accurate elevation modeling for off-road navigation via neural processes,"

TABLE S13
INTERFACE CANDIDATES RANKED BY THE RULE FIXED IN CODEBOOK VERSION 1.0 BEFORE CODING. A CANDIDATE IS A FUNCTION PAIR WITH ONE COUPLING SOURCE. N COUNTS ITS WORKS, J THE WORKS WITH DIRECT JOINT WHEELED EVIDENCE AND W THE TARGET WHEELED WORKS. RANK A DROPS THE INTEGRATION-QUESTIONS STAGE AND RANK B SORTS BY GAP SHARE.

Rank	Candidate	n	j	w	Gap	Rank A	Rank B	Status	Reason
1	LOC-MRC:COM	17	0	12	17	1	12	Interface 4	
2	PLN-MRC:COM	15	1	12	14	2	16	rejected	Merged into Interface 4. Rendezvous, reception-aware control and communication-aware allocation already plan around links (Sec. VII-C). The open part is acceptance of information after reconnection.
3	LOC-MRC:SFA	9	0	6	9	3	13	rejected	Merged into Interface 4. Outlier loop closures and faulty shared landmarks are the corrupted content that its comparison injects as landmark bias.
4	LOC-REL:PER	9	1	5	8	4	17	Interface 1	
5	LOC-REL:SFA	7	0	3	7	7	5	Interface 1	
6	PLN-SEM:DYN	10	4	8	6	6	25	rejected	Merged into Interface 2, whose comparison varies obstacle-track quality. Four works already give direct joint wheeled evidence for social or semantic choices among moving people.
7	MRC-REL:SFA	6	0	3	6	13	7	rejected	Merged into Interface 4. Fault propagation through shared state is one of its primary outcomes.
8	LOC-MRC:PER	6	0	5	6	5	14	rejected	Merged into Interface 4. Teammate ranges and maps serve as complementary sensing (Sec. IV-B), and their acceptance after a partition is the interface question.
9	PLN-REL:ACT	5	0	5	5	12	1	rejected	Named as a further gap in Sec. X-E. The five works accommodate modeled actuator faults inside one tracking controller, without a response chosen across functions.
10	PLN-SEM:MOD	7	3	6	4	8	26	Interface 2	
11	LOC-PLN:PER	5	1	5	4	9	18	Interface 3	
26	PLN-SEM:RES	6	4	6	2	29	29	Interface 2	
–	LOC-PLN:TER	2	0	1	2	–	–	Interface 3	
–	SEM-REL:MOD	1	0	0	1	–	–	Interface 2	
–	SEM-REL:RES	0	0	0	0	–	–	Interface 2	

in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, J. Lim, S. Song, and H.-W. Park, Eds., vol. 305. PMLR, 27–30 Sep 2025, pp. 2802–2822. [Online]. Available: https://proceedings.mlr.press/v305/jung25b.html

[38] Q. Bi, X. Zhang, S. Zhang, R. Wang, L. Li, and J. Yuan, "LRAE: Large-region-aware safe and fast autonomous exploration of ground robots for uneven terrains," *IEEE Robotics and Automation Letters*, vol. 9, no. 12, pp. 11 186–11 193, 2024.

[39] S. Xie, R. Song, Y. Zhao, X. Huang, Y. Li, and W. Zhang, "Circular accessible depth: A robust traversability representation for UGV navigation," *IEEE Transactions on Robotics*, vol. 39, no. 6, pp. 4875–4891, 2023.

[40] T. Guan, D. Kothandaraman, R. Chandra, A. J. Sathyamoorthy, K. Weerakoon, and D. Manocha, "GA-Nav: Efficient terrain segmentation for robot navigation in unstructured outdoor environments," *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 8138–8145, 2022.

[41] Y. Zhao, L. Zhou, C. Zhang, and Y. Li, "Anisotropic terrain-aware autonomous exploration via image-domain perception and sparse routing," *IEEE Robotics and Automation Letters*, vol. 11, no. 10, pp. 11 062–11 069, 2026.

[42] R. Wang *et al.*, "Vehicle-cloud cooperative trajectory planning via switched model predictive control for UGVs in unstructured transportation environment," *IEEE Transactions on Intelligent Transportation Systems*, vol. 27, no. 8, pp. 9517–9528, 2026.

[43] S. Keyumarsi, M. W. S. Atman, and A. Gusrialdi, "Circulation-embedded control barrier function for safe navigation: A solution to avoid undesired equilibria and dysfunctional circulation," *Robotics and Autonomous Systems*, vol. 194, p. 105132, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0921889025002295

[44] R. Das, S. Upadhyay, and P. Jagtap, "Real-time spatiotemporal tubes for dynamic unsafe sets," *IEEE Robotics and Automation Letters*, vol. 11, no. 2, pp. 2146–2153, 2026.

[45] A. Dixit, Z. Mei, M. Booker, M. Storey-Matsutani, A. Z. Ren, and A. Majumdar, "Perceive with confidence: Statistical safety assurances for navigation with learning-based perception," in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, P. Agrawal, O. Kroemer, and W. Burgard, Eds., vol. 270. PMLR, 06–09 Nov 2025, pp. 2517–2541. [Online]. Available: https://proceedings.mlr.press/v270/dixit25a.html

[46] O. de Groot, L. Ferranti, D. M. Gavrila, and J. Alonso-Mora, "Topology-driven parallel trajectory optimization in dynamic environments," *IEEE Transactions on Robotics*, vol. 41, pp. 110–126, 2025.

[47] S. Triest, M. Sivaprakasam, S. Aich, D. Fan, W. Wang, and S. Scherer, "Velociraptor: Leveraging visual foundation models for label-free, risk-aware off-road navigation," in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, P. Agrawal, O. Kroemer, and W. Burgard, Eds., vol. 270. PMLR, 06–09 Nov 2025, pp. 4483–4494. [Online]. Available: https://proceedings.mlr.press/v270/triest25a.html

[48] Z. Dong *et al.*, "Learning smooth state-dependent traversability from dense point clouds," in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, J. Lim, S. Song, and H.-W. Park, Eds., vol. 305. PMLR, 27–30 Sep 2025, pp. 1654–1671. [Online]. Available: https://proceedings.mlr.press/v305/dong25a.html

[49] D. Song, J. Liang, A. Payandeh, A. H. Raj, X. Xiao, and D. Manocha, "VLM-Social-Nav: Socially aware robot navigation through scoring using vision-language models," *IEEE Robotics and Automation Letters*, vol. 10, no. 1, pp. 508–515, 2025.

[50] B. Yu, Q. Yuan, K. Li, H. Kasaei, and M. Cao, "Co-NavGPT: Multi-robot cooperative visual semantic navigation using vision language models," *IEEE Robotics and Automation Letters*, vol. 11, no. 2, pp. 2122–2129, 2026.

[51] Y. Tang *et al.*, "OpenIN: Open-vocabulary instance-oriented navigation in dynamic domestic environments," *IEEE Robotics and Automation Letters*, vol. 10, no. 9, pp. 9256–9263, 2025.

[52] N. Yokoyama, S. Ha, D. Batra, J. Wang, and B. Bucher, "VLFM: Vision-language frontier maps for zero-shot semantic navigation," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 42–48.

[53] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard, "Hierarchical open-vocabulary 3D scene graphs for language-grounded robot navigation," in *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024.

[54] J. Zhang *et al.*, "Uni-NaVid: A video-based vision-language-action model for unifying embodied navigation tasks," in *Proceedings of Robotics: Science and Systems*, LosAngeles, CA, USA, June 2025.

[55] D. Song, J. Liang, X. Xiao, and D. Manocha, "VL-TGS: Trajectory generation and selection using vision language models in mapless outdoor environments," *IEEE Robotics and Automation Letters*, vol. 10, no. 6, pp. 5791–5798, 2025.

[56] S. Luo *et al.*, "GSON: A group-based social navigation framework with large multimodal model," *IEEE Robotics and Automation Letters*, vol. 10, no. 10, pp. 9646–9653, 2025.

[57] M. Elnoor *et al.*, "VLM-GroNav: Robot navigation using physically grounded vision-language models in outdoor environments," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 2391–2398.

[58] M. J. Munje *et al.*, "SocialNav-SUB: Benchmarking VLMs for scene understanding in social robot navigation," in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, J. Lim, S. Song, and H.-W. Park, Eds., vol. 305. PMLR, 27–30 Sep 2025, pp. 1120–1143. [Online]. Available: https://proceedings.mlr.press/v305/munje25a.html

[59] A. Asgharivaskasi, F. Girke, and N. Atanasov, "Riemannian optimization for active mapping with robot teams," *IEEE Transactions on Robotics*, vol. 41, pp. 1077–1097, 2025.

[60] R. Bai, S. Yuan, H. Guo, P. Yin, W.-Y. Yau, and L. Xie, "Multi-robot active graph exploration with reduced Pose-SLAM uncertainty via submodular optimization," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 10 229–10 236.

[61] V. Vijay, K. A. Pant, M. Cho, Y. Guo, J. M. Goppert, and I. Hwang, "Range-based multi-robot integrity monitoring for cyberattacks and faults: An anchor-free approach," *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2630–2637, 2025.

[62] H. Kim *et al.*, "Commerge: Communication-efficient, robust, and fast LiDAR map merging framework for multi-robot coordination in resource-constrained scenarios," *The International Journal of Robotics Research*, vol. 0, no. 0, p. 02783649261464202, 2026. [Online]. Available: https://doi.org/10.1177/02783649261464202

[63] D. McGann and M. Kaess, "iMESA: Incremental distributed optimization for collaborative simultaneous localization and mapping," in *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024.

[64] M. Escourrou, J. Al Hage, and P. Bonnifait, "Fault tolerant decentralized collaboration for simultaneous localization and prior map update with stable 2D features," *Robotics and Autonomous Systems*, vol. 200, p. 105402, 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0921889026000758

[65] P. Li, J. Liu, Y. Wu, and L. Zhou, "Failure-aware multi-robot coordination for resilient and adaptive target tracking," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 4023–4038, 2026.

[66] J. Bayer and J. Faigl, "Decentralized multi-robot exploration under low-bandwidth communications," *Autonomous Robots*, vol. 50, no. 1, p. 7, Dec 2025. [Online]. Available: https://doi.org/10.1007/s10514-025-10234-3

[67] Y.-H. Lee, C. Zhu, T. Wiedemann, E. Staudinger, S. Zhang, and C. Günther, "CoVOR-SLAM: Cooperative SLAM using visual odometry and ranges for multi-robot systems," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 7, pp. 9358–9369, 2025.

[68] P.-Y. Lajoie and G. Beltrame, "Swarm-SLAM: Sparse decentralized collaborative simultaneous localization and mapping framework for multi-robot systems," *IEEE Robotics and Automation Letters*, vol. 9, no. 1, pp. 475–482, 2024.

[69] P. Zhong, F. Rossi, and D. A. Shell, "Planned synchronization for multi-robot systems with active observations," *Autonomous Robots*, vol. 50, no. 1, 2026, article 5; first published online 24 December 2025; assigned to the March 2026 issue. [Online]. Available: https://doi.org/10.1007/s10514-025-10225-4

[70] W. Zhang, L. Cheng, J. Wei, X. Xu, Z. Zhang, and B. Xin, "Bandwidth-efficient collaborative SLAM: Mutual information for data selection and finite-state entropy for lossless compression," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 5401–5412, 2026.

[71] M. H. Shahna, P. Mustalahti, and J. Mattila, "NMPC-augmented visual navigation and safe learning control for large-scale mobile robots," *IEEE Robotics and Automation Letters*, vol. 11, no. 4, pp. 5182–5189, 2026.

[72] S. Zhang, V. Puig, and S. Ifqir, "Robust LPV fault diagnosis using the set-based approach for autonomous ground vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 8, pp. 9078–9090, 2024.

[73] N. Akai, "Reliable Monte Carlo localization for mobile robots," *Journal of Field Robotics*, vol. 40, no. 3, pp. 595–613, 2023. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.22149

[74] S. Verma and D. S. Bernstein, "Real-time kinematics-based sensor-fault detection for autonomous vehicles using single and double transport with adaptive numerical differentiation," *Control Engineering Practice*, vol. 159, p. 106304, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S096706612500067X

[75] A. De Luca, L. Muratore, and N. G. Tsagarakis, "Autonomous navigation with online replanning and recovery behaviors for wheeled-legged robots using behavior trees," *IEEE Robotics and Automation Letters*, vol. 8, no. 10, pp. 6803–6810, 2023.

[76] X. Zheng, W. Wen, and L.-T. Hsu, "Safety-quantifiable line feature-based monocular visual localization with 3D prior map," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 7, pp. 9226–9240, 2025.

[77] R. Ma, Y. Yu, S. Zhang, and P. Shi, "Event-triggered fault-tolerant tracking control for autonomous ground vehicle via descriptor reduced-order observer," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 7, pp. 10 308–10 319, 2025.

[78] C. Agia *et al.*, "Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress," in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, P. Agrawal, O. Kroemer, and W. Burgard, Eds., vol. 270. PMLR, 06–09 Nov 2025, pp. 689–723. [Online]. Available: https://proceedings.mlr.press/v270/agia25a.html

[79] G. Kulathunga *et al.*, "Resilient timed elastic band planner for collision-free navigation in unknown environments," *Journal of Field Robotics*, vol. 42, no. 7, pp. 3902–3917, 2025. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.22602

[80] C. Chen, Y. Liu, Y. Zhuang, S. Mao, K. Xue, and S. Zhou, "SCALE: Self-correcting visual navigation for mobile robots via anti-novelty estimation," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 16 360–16 366.

[81] K. Stachowicz, L. Ignatova, and S. Levine, "Lifelong autonomous improvement of navigation foundation models in the wild," in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, P. Agrawal, O. Kroemer, and W. Burgard, Eds., vol. 270. PMLR, 06–09 Nov 2025, pp. 1035–1047. [Online]. Available: https://proceedings.mlr.press/v270/stachowicz25a.html

[82] J. Wang, C. Cai, Y. Wang, Y. Li, S. Zhang, and H. Chen, "Screw-based feature constraint model and degeneracy analysis for robotic state estimation: Theory and experiments," *The International Journal of Robotics Research*, vol. 0, no. 0, p. 02783649261463720, 2026. [Online]. Available: https://doi.org/10.1177/02783649261463720

[83] A. Tao, Y. Luo, C. Guo, C. Xia, M. Zhao, and X. Li, "An equivariant filter-based state estimator for tightly-coupled LiDAR–radar–inertial odometry," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 14 926–14 940, 2026.

[84] T. Okawara *et al.*, "Tightly-coupled LiDAR-IMU-wheel odometry with an online neural kinematic model learning via factor graph optimization," *Robotics and Autonomous Systems*, vol. 187, p. 104929, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0921889025000156

[85] S. Zhao *et al.*, "Resilient odometry via hierarchical adaptation," *Science Robotics*, vol. 10, no. 109, p. eadv1818, 2025. [Online]. Available: https://www.science.org/doi/abs/10.1126/scirobotics.adv1818

[86] Z. Zhang, W. Bai, L. Zhao, and P. Ladosz, "PL-VIWO: A lightweight and robust point-line monocular visual inertial wheel odometry," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 15 205–15 212.

[87] M. Nissov, N. Khedekar, and K. Alexis, "Degradation resilient LiDAR-radar-inertial odometry," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 8587–8594.

[88] W. Zhou, Y. Pan, J. Liu, T. Wang, and H. Zha, "Visual-inertial-wheel odometry with wheel-aided maximum-a-posteriori initialization for ground robots," *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 4814–4821, 2024.

[89] Y. Xie, Y. Guo, Z. Mi, X. Wang, Y. Yang, and M. S. Obaidat, "Indoor visual re-localization for long-term autonomous robots based on object-level features and semantic relationships," *IEEE Robotics and Automation Letters*, vol. 8, no. 2, pp. 840–847, 2023.

[90] J. Yin, A. Li, T. Li, W. Yu, and D. Zou, "M2DGR: A multi-sensor and multi-scenario SLAM dataset for ground robots," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 2266–2273, 2022.

[91] J. Morris, Y. Wang, M. Kliniewski, and V. Ila, "DynoSAM: Open-source smoothing and mapping framework for dynamic SLAM," *IEEE Transactions on Robotics*, vol. 42, pp. 1178–1197, 2026.

[92] Z. Wang *et al.*, "LiteFT-PR: Lightweight and fault-tolerant LiDAR-camera fusion network for robust place recognition via model distillation," *IEEE Robotics and Automation Letters*, vol. 11, no. 4, pp. 4267–4274, 2026.

[93] Z. Huang, C. Sun, J. Leng, and B. Wang, "AESF-LIO: Adaptive error-state fusion LiDAR-inertial odometry for ground vehicles in structured environments," *IEEE Robotics and Automation Letters*, vol. 11, no. 6, pp. 7372–7379, 2026.

[94] J. Li, J. Ji, Q. Wang, H. Yu, Y. Pan, and F. Gao, "Active collision-based navigation for wheeled robots," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 10 932–10 938.

[95] F. Xie and S. Schwertfeger, "Robust lifelong indoor LiDAR localization using the area graph," *IEEE Robotics and Automation Letters*, vol. 9, no. 1, pp. 531–538, 2024.

[96] F. Crocetti, R. Brilli, A. Dionigi, M. L. Fravolini, G. Costante, and P. Valigi, "Comparison of DSO and ORB-SLAM3 in low-light environments with auxiliary lighting and deep learning based image enhancing," *Journal of Field Robotics*, vol. 42, no. 7, pp. 3748–3771, 2025. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.22595

[97] J. Ruan and D. Zhang, "HK-MEMS, a multi-sensor data set with MEMS LiDAR on degenerate and dynamic urban scenarios," *Journal of Field Robotics*, vol. 43, no. 3, pp. 2159–2182, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.70136

[98] L. Zhang, Y. Tao, J. Lin, F. Zhang, and M. Fallon, "Visual localization in 3D maps: comparing point cloud, mesh, and NeRF representations," *Autonomous Robots*, vol. 50, no. 1, 2026, article 14. [Online]. Available: https://doi.org/10.1007/s10514-025-10232-5

[99] Y. Dong, E. Xu, J. Yang, S. Qiu, J. Wang, and B. Han, "Implicit point-to-voxel LiDAR-IMU SLAM," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 8097–8110, 2026.

[100] Z. Liao *et al.*, "A real-time degeneracy sensing and compensation method for enhanced LiDAR SLAM," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 3, pp. 4202–4213, 2025.

[101] B. Song, X. Xia, P. Yan, Y. Zhong, W. Wen, and L.-T. Hsu, "Online IMU-odometer calibration using GNSS measurements for autonomous ground vehicle localization," *IEEE Transactions on Intelligent Transportation Systems*, vol. 27, no. 10, pp. 12 289–12 302, 2026.

[102] Z. Chen *et al.*, "Heterogeneous LiDAR dataset for benchmarking robust localization in diverse degenerate scenarios," *The International Journal of Robotics Research*, vol. 45, no. 1, pp. 6–22, 2026. [Online]. Available: https://doi.org/10.1177/02783649251344967

[103] W. Yang, D. Lee, M. Jung, and A. Kim, "Geometrically-constrained radar-inertial odometry via continuous point-pose uncertainty modeling," *IEEE Robotics and Automation Letters*, vol. 11, no. 5, pp. 6098–6105, 2026.

[104] Z. Jin and C. Jiang, "Range-aided LiDAR-inertial multi-vehicle localization and mapping in degenerate environments," *IEEE Transactions on Intelligent Vehicles*, vol. 11, no. 1, pp. 1–14, 2026.

[105] S. Wang, F. Cao, T. Wang, S. Shao, and L. Liu, "Robust ground constrained SLAM for mobile robot with sparse-channel LiDAR," *IEEE Transactions on Intelligent Vehicles*, vol. 10, no. 5, pp. 3291–3302, 2025.

[106] K. Xu, Y. Hao, S. Yuan, C. Wang, and L. Xie, "AirSLAM: An efficient and illumination-robust point-line visual SLAM system," *IEEE Transactions on Robotics*, vol. 41, pp. 1673–1692, 2025.

[107] J. Kim, M. Jeon, J. Min, K. Kwak, and J. Seo, "E2-BKI: Evidential ellipsoidal Bayesian kernel inference for uncertainty-aware Gaussian semantic mapping," *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 2378–2385, 2026.

[108] R. Xiao *et al.*, "LiDAR teach, radar repeat: Robust cross-modal navigation in degenerate and varying environments," *IEEE Transactions on Robotics*, vol. 42, pp. 2500–2520, 2026.

[109] Z. Li, W. Wang, Q. Shen, Y. Zhao, and Z. Fang, "Super4DR: 4-d radar-centric self-supervised odometry and Gaussian-based map optimization," *IEEE Transactions on Robotics*, vol. 42, pp. 3256–3276, 2026.

[110] Y. Xu, R. Zheng, S. Zhang, M. Liu, and J. Yu, "Uncertainty-aware autonomous robot exploration using confidence-rich localization and mapping," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 1124–1138, 2025.

[111] K. Long, Y. Yi, Z. Dai, S. Herbert, J. Cortés, and N. Atanasov, "Sensor-based distributionally robust control for safe robot navigation in dynamic environments," *The International Journal of Robotics Research*, vol. 45, no. 2, pp. 328–351, 2026. [Online]. Available: https://doi.org/10.1177/02783649251352000

[112] T. Zeng, H. Ye, J. Peng, J. Yu, H. Chen, and H. Zhang, "TARIC: Memory-augmented traversability-aware outdoor VLN under interrupted semantic cues," *IEEE Robotics and Automation Letters*, vol. 11, no. 9, pp. 10 082–10 089, 2026.

[113] J. Wang, D. Huo, Y. Shi, C. Gao, Y. Qiao, and G. Zhou, "OPEN: Lightweight map-based semantic navigation for GPS-free last-mile delivery," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 20 713–20 725, 2025.

[114] Y. Sun *et al.*, "PROMO-Orchard: A robust semantic-enhanced framework for dual-robot simultaneous localization and mapping merging in orchard," *Journal of Field Robotics*, vol. 43, no. 7, pp. 4880–4903, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.70323

[115] J. Liu and G. Hu, "Extended range of localization services via formation-controlled mobile ultra-wideband anchors," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 265–277, 2026.

[116] J. Wang, C. Cai, J. Tan, and H. Chen, "Communication-robust asynchronous distributed LiDAR collaborative smoothing and mapping," *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 3700–3707, 2026.

[117] D. Milojevic, G. Zardini, M. Elser, A. Censi, and E. Frazzoli, "CODEI: Resource-efficient task-driven co-design of perception and decision making for mobile robots applied to autonomous vehicles," *IEEE Transactions on Robotics*, vol. 41, pp. 2727–2748, 2025.

[118] H. Huang, M. Zhang, J. Hu, and X. Guan, "Unveiling non-reproducibility in LiDAR-inertial odometry," *IEEE Robotics and Automation Letters*, vol. 11, no. 1, pp. 994–1001, 2026.

[119] A. Paudel, X. Xiao, and G. J. Stein, "Multi-strategy deployment-time learning and adaptation for navigation under uncertainty," in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, P. Agrawal, O. Kroemer, and W. Burgard, Eds., vol. 270. PMLR, 06–09 Nov 2025, pp. 3908–3923. [Online]. Available: https://proceedings.mlr.press/v270/paudel25a.html

[120] F. Schmidt, C. Blessing, M. Enzweiler, and A. Valada, "Visual-inertial SLAM for unstructured outdoor environments: Benchmarking the benefits and computational costs of loop closing," *Journal of Field Robotics*, vol. 42, no. 7, pp. 3726–3747, 2025. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.22581

[121] K. Naveed, W. Hussain, I. Hussain, D. Lee, and M. L. Anjum, "Help me through: Imitation learning based active view planning to avoid SLAM tracking failures," *IEEE Transactions on Robotics*, vol. 41, pp. 4236–4252, 2025.

[122] M. Zarei and R. Chhabra, "Fault-tolerant multimodal localization of multirobots on matrix Lie groups," *IEEE Transactions on Robotics*, vol. 42, pp. 2459–2477, 2026.

[123] Y. Ma *et al.*, "NVMS-SLAM: Normal vector-based multi-session LiDAR SLAM in indoor environments," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 8034–8045, 2026.

[124] S. Gupta, T. Guadagnino, B. Mersch, N. Trekel, M. V. R. Malladi, and C. Stachniss, "Efficiently closing loops in LiDAR-based SLAM using point cloud density maps," *The International Journal of Robotics Research*, vol. 0, no. 0, p. 02783649261449269, 2026. [Online]. Available: https://doi.org/10.1177/02783649261449269

[125] Y. Zhang *et al.*, "BIM-Loc: BIM-integrated discrepancy-aware LiDAR-based indoor localization," *The International Journal of Robotics Research*, vol. 0, no. 0, p. 02783649261462593, 2026. [Online]. Available: https://doi.org/10.1177/02783649261462593

[126] Z. Hu, R. Ma, S. Zhang, H. Wang, X. Liu, and D. Zhao, "Resilient memory-event-triggered predictive tracking control for unmanned ground vehicle," *IEEE Transactions on Intelligent Vehicles*, vol. 11, no. 4, pp. 500–510, 2026.

[127] P.-Y. Lajoie, B. Ramtoula, D. De Martini, and G. Beltrame, "3D foundation model-based loop closing for decentralized collaborative SLAM," *IEEE Robotics and Automation Letters*, vol. 10, no. 11, pp. 11 188–11 195, 2025.

[128] D. McGann and M. Kaess, "riMESA: Consensus ADMM for real-world collaborative SLAM," *IEEE Transactions on Robotics*, vol. 42, pp. 3550–3569, 2026.

[129] O. A. Hafez, M. Joerger, and M. Spenko, "Quantifying mobile robot localization safety for an EKF-based SLAM estimator: An integrity monitoring approach," *The International Journal of Robotics Research*, vol. 44, no. 6, pp. 972–988, 2025. [Online]. Available: https://doi.org/10.1177/02783649241287797

[130] ——, "How safe is particle filtering-based localization for mobile robots? an integrity monitoring approach," *IEEE Transactions on Robotics*, vol. 40, pp. 3372–3387, 2024.
[131] J.-X. Zhang, J. Zhao, and P. K. Wong, "Fault-tolerant trajectory tracking for unmanned ground vehicles via virtual error-based prescribed performance control," *IEEE Transactions on Automatic Control*, vol. 71, no. 4, pp. 2660–2667, 2026.
[132] H. Zhang, Z. Li, and A. Clark, "Safe control for nonlinear systems under faults and attacks via control barrier functions," *IEEE Transactions on Automatic Control*, vol. 70, no. 11, pp. 7334–7349, 2025.
[133] J. O'Keeffe, "Anticipating degradation: A predictive approach to fault tolerance in robot swarms," *IEEE Robotics and Automation Letters*, vol. 10, no. 9, pp. 8954–8961, 2025.
[134] G. Notomista *et al.*, "A resilient and energy-aware task allocation framework for heterogeneous multirobot systems," *IEEE Transactions on Robotics*, vol. 38, no. 1, pp. 159–179, 2022.
[135] E. Hashemi, X. He, and K. H. Johansson, "Robust slip-aware fusion for mobile robots state estimation," *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 7896–7903, 2022.
[136] M. Zhang, X. Zuo, Y. Chen, Y. Liu, and M. Li, "Pose estimation for ground robots: On manifold representation, integration, reparameterization, and optimization," *IEEE Transactions on Robotics*, vol. 37, no. 4, pp. 1081–1099, 2021.
[137] H. Gao, Q. Qiu, H. Liu, D. Liang, C. Wang, and X. Zhang, "ERPoT: Effective and reliable pose tracking for mobile robots using lightweight polygon maps," *IEEE Transactions on Robotics*, vol. 41, pp. 3799–3819, 2025.
[138] N. Li, J. Guo, L. Ding, C. Tian, C. Zhou, and H. Gao, "Estimation of slip ratio and side slip angle of wheeled planetary rovers based on trace imprint," *IEEE Robotics and Automation Letters*, vol. 10, no. 8, pp. 7979–7986, 2025.
[139] C. Jiang, K. Zhang, S. Yang, S. Shen, C. Xu, and F. Gao, "WING: Wheel-inertial neural odometry with ground manifold constraints," *IEEE Transactions on Intelligent Vehicles*, vol. 10, no. 3, pp. 1510–1525, 2025.
[140] H. Hu, K. Zhang, A. H. Tan, M. Ruan, C. Agia, and G. Nejat, "A sim-to-real pipeline for deep reinforcement learning for autonomous robot navigation in cluttered rough terrain," *IEEE Robotics and Automation Letters*, vol. 6, no. 4, pp. 6569–6576, 2021.
[141] P. Sherman and N. Bezzo, "A unified framework for mission-gain-aware proactive planning and control of autonomous mobile robots," *IEEE Robotics and Automation Letters*, vol. 11, no. 10, pp. 12 192–12 199, 2026.
[142] E. Daş and J. W. Burdick, "Robust control barrier functions using uncertainty estimation with application to mobile robots," *IEEE Transactions on Automatic Control*, vol. 70, no. 7, pp. 4766–4773, 2025.
[143] Y. Li *et al.*, "Real-time multilevel terrain-aware path planning for ground mobile robots in large-scale rough terrains," *IEEE Transactions on Robotics*, vol. 41, pp. 4159–4179, 2025.
[144] Z. Jian, Z. Liu, H. Shao, X. Wang, X. Chen, and B. Liang, "Path generation for wheeled robots autonomous navigation on vegetated terrain," *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1764–1771, 2024.
[145] J. Wang, M. T. H. Fader, and J. A. Marshall, "Learning-based model predictive control for improved mobile robot path following using Gaussian processes and feedback linearization," *Journal of Field Robotics*, vol. 40, no. 5, pp. 1014–1033, 2023. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.22165
[146] I. Cho and W. Chung, "Learning self-supervised traversability with navigation experiences of mobile robots: A risk-aware self-training approach," *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 4122–4129, 2024.
[147] O. Jerome, G. Kulathunga, and T. Nascimento, "A terrain-adaptive ϵ-constraint MPC for uneven terrain kinodynamic planning," *IEEE Robotics and Automation Letters*, vol. 11, no. 10, pp. 11 865–11 872, 2026.
[148] Y. Jia, Q. Li, and J. P. How, "Off-road navigation via implicit neural representation of terrain traversability," *IEEE Robotics and Automation Letters*, vol. 11, no. 6, pp. 7126–7133, 2026.
[149] M. G. Castro *et al.*, "VAMOS: A hierarchical vision-language-action model for capability-modulated and steerable navigation," *IEEE Robotics and Automation Letters*, vol. 11, no. 11, pp. 13 055–13 062, 2026.
[150] A. H. Tan, S. Narasimhan, and G. Nejat, "4CNet: A diffusion approach to map prediction for decentralized multirobot exploration," *IEEE Transactions on Robotics*, vol. 42, pp. 1371–1388, 2026.
[151] T. Ji, A. N. Sivakumar, G. Chowdhary, and K. Driggs-Campbell, "Proactive anomaly detection for robot navigation with multi-sensor fusion," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4975–4982, 2022.
[152] N. Hirose, D. Shah, K. Stachowicz, A. Sridhar, and S. Levine, "SELFI: Autonomous self-improvement with RL for vision-based navigation around people," in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, P. Agrawal, O. Kroemer, and W. Burgard, Eds., vol. 270. PMLR, 06–09 Nov 2025, pp. 97–116. [Online]. Available: https://proceedings.mlr.press/v270/hirose25a.html
[153] F.-x. Chen *et al.*, "FusedNet: End-to-end mobile robot relocalization in dynamic large-scale scene," *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 4099–4105, 2024.
[154] H. Li, X. Meng, X. Zuo, Z. Liu, H. Wang, and D. Cremers, "PG-SLAM: Photorealistic and geometry-aware RGB-D SLAM in dynamic environments," *IEEE Transactions on Robotics*, vol. 41, pp. 6084–6101, 2025.
[155] H. Gil, D. Lee, G. Kim, and A. Kim, "Ephemerality meets lidar-based lifelong mapping," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 3312–3319.
[156] Y. Zhang, M. Bujanca, and M. Luján, "NGD-SLAM: Towards real-time dynamic SLAM without GPU," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 3467–3473.
[157] K. Huang, W. Yang, P. Zhou, L. Li, and J. Yao, "JPG-SLAM: Joint Point-Gaussian splatting representation for dense dynamic SLAM," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 5018–5024.
[158] L. Yang, S. M. Prakhya, S. Zhu, and Z. Liu, "Lifelong 3D mapping framework for hand-held & robot-mounted LiDAR mapping systems," *IEEE Robotics and Automation Letters*, vol. 9, no. 11, pp. 9446–9453, 2024.
[159] B. Fang, G. Mei, X. Yuan, L. Wang, Z. Wang, and J. Wang, "Visual SLAM for robot navigation in healthcare facility," *Pattern Recognition*, vol. 113, p. 107822, 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0031320321000091
[160] D. C. Herraez, M. Zeller, D. Wang, J. Behley, M. Heidingsfeld, and C. Stachniss, "Multi-session mapping and long-term localization for autonomous vehicles using radar," *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 2522–2529, 2026.
[161] Y. Li *et al.*, "FRTree planner: Robot navigation in cluttered and unknown environments with tree of free regions," *IEEE Robotics and Automation Letters*, vol. 10, no. 4, pp. 3811–3818, 2025.
[162] J. de Heuvel, X. Zeng, W. Shi, T. Sethuraman, and M. Bennewitz, "Spatiotemporal attention enhances lidar-based robot navigation in dynamic environments," *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 4202–4209, 2024.
[163] Z. Zhang, H. Hajieghrary, E. Dean, and K. Åkesson, "Prescient collision-free navigation of mobile robots with iterative multimodal motion prediction of dynamic obstacles," *IEEE Robotics and Automation Letters*, vol. 8, no. 9, pp. 5488–5495, 2023.
[164] F. Vesentini, D. Meli, N. Sansonetto, L. D. Persio, and R. Muradore, "Dynamic movement primitives with control barrier functions for constrained trajectory planning," *IEEE Robotics and Automation Letters*, vol. 10, no. 9, pp. 9416–9421, 2025.
[165] X. Wang, T. Kim, B. Hoxha, G. Fainekos, and D. Panagou, "Safe navigation in uncertain crowded environments using risk adaptive CVaR barrier functions," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 7669–7676.
[166] Y. Tao, M. Li, X. Cao, and P. Lu, "Mobile robot collision avoidance based on deep reinforcement learning with motion constraints," *IEEE Transactions on Intelligent Vehicles*, vol. 10, no. 3, pp. 2163–2173, 2025.
[167] Z. Sun, B. Xia, P. Xie, X. Li, and J. Wang, "NAMR-RRT: Neural adaptive motion planning for mobile robots in dynamic environments," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 13 087–13 100, 2025.
[168] X. Kong, W. Ning, Y. Xia, Z. Sun, and H. Xie, "Adaptive high-order control barrier function-based iterative LQR for real time safety-critical motion planning," *IEEE Robotics and Automation Letters*, vol. 9, no. 7, pp. 6099–6106, 2024.
[169] S. Keyumarsi, M. W. S. Atman, and A. Gusrialdi, "LiDAR-based online control barrier function synthesis for safe navigation in unknown environments," *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1043–1050, 2024.

[170] Z. Liu, W. Na, C. Yao, C. Liu, and Q. Chen, "Relaxing the limitations of the optimal reciprocal collision avoidance algorithm for mobile robots in crowds," *IEEE Robotics and Automation Letters*, vol. 9, no. 6, pp. 5520–5527, 2024.

[171] J. Cao, S. Khan, W. Liu, Y. Li, and B. Vucetic, "Remote UGV control via practical wireless channels: A model predictive control approach," *IEEE Transactions on Intelligent Vehicles*, pp. 1–13, 2024.

[172] A. Hierholz, V. Reß, N. Hentschel, A. Gienger, N. Haala, and O. Sawodny, "Active visual SLAM using potential fields and model predictive control," *Control Engineering Practice*, vol. 172, p. 106897, 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0967066126001413

[173] J. Yao, X. Zhang, Y. Xia, Z. Wang, A. Roy-Chowdhury, and J. Li, "Towards generalizable safety in crowd navigation via conformal uncertainty handling," in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, J. Lim, S. Song, and H.-W. Park, Eds., vol. 305. PMLR, 27–30 Sep 2025, pp. 4206–4225. [Online]. Available: https://proceedings.mlr.press/v305/yao25a.html

[174] S. Liu *et al.*, "HEIGHT: Heterogeneous interaction graph transformer for robot navigation in crowded and constrained environments," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 1211–1230, 2026.

[175] Z. Xie and P. Dames, "SCOPE: Stochastic cartographic occupancy prediction engine for uncertainty-aware dynamic navigation," *IEEE Transactions on Robotics*, vol. 41, pp. 4139–4158, 2025.

[176] Z. Fang, A. Xiao, D. Hsu, and G. H. Lee, "From obstacles to etiquette: Robot social navigation with VLM-informed path selection," *IEEE Robotics and Automation Letters*, vol. 11, no. 4, pp. 3947–3954, 2026.

[177] H. Duan *et al.*, "CAUSALNAV: A long-term embodied navigation system for autonomous mobile robots in dynamic outdoor scenarios," *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 3198–3205, 2026.

[178] L. Yue, D. Zhou, L. Xie, F. Zhang, Y. Yan, and E. Yin, "Safe-VLN: Collision avoidance for vision-and-language navigation of autonomous robots operating in continuous environments," *IEEE Robotics and Automation Letters*, vol. 9, no. 6, pp. 4918–4925, 2024.

[179] S. Narasimhan, A. H. Tan, D. Choi, and G. Nejat, "OLiVia-Nav: An online lifelong vision language approach for mobile robot social navigation," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 9130–9137.

[180] M. Ganai, R. Sinha, C. Agia, D. Morton, L. D. Lillo, and M. Pavone, "Real-time out-of-distribution failure prevention via multi-modal reasoning," in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, J. Lim, S. Song, and H.-W. Park, Eds., vol. 305. PMLR, 27–30 Sep 2025, pp. 283–308. [Online]. Available: https://proceedings.mlr.press/v305/ganai25a.html

[181] A. T. Buyukkocak and D. Aksaray, "Resilient online planning for mobile robots with minimal relaxation of signal temporal logic specifications," *IEEE Robotics and Automation Letters*, vol. 10, no. 6, pp. 5935–5942, 2025.

[182] J. Kim, J. Sim, W. Kim, K. P. Sycara, and C. Nam, "CARE: Enhancing safety of visual navigation through collision avoidance via repulsive estimation," in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, J. Lim, S. Song, and H.-W. Park, Eds., vol. 305. PMLR, 27–30 Sep 2025, pp. 3704–3719. [Online]. Available: https://proceedings.mlr.press/v305/kim25c.html

[183] C. Dawson, A. Parashar, and C. Fan, "RADIUM: Predicting and repairing end-to-end robot failures using gradient-accelerated sampling," *IEEE Transactions on Robotics*, vol. 41, pp. 2268–2284, 2025.

[184] E. Liu, W. Tang, A. Huang, Z. Zhou, and R. Zhang, "A season-robust long-term localization method using trunk semantic features in dynamic orchard environments," *Journal of Field Robotics*, vol. 43, no. 4, pp. 2725–2746, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.70159

[185] J. Fu, Y. Du, K. Singh, J. B. Tenenbaum, and J. J. Leonard, "NeuSE: Neural SE(3)-equivariant embedding for long-term object-based simultaneous localization and mapping," *The International Journal of Robotics Research*, vol. 45, no. 1, pp. 159–189, 2026. [Online]. Available: https://doi.org/10.1177/02783649251355966

[186] S. Jang, A. J. Lee, I. M. A. Nahrendra, and H. Myung, "Chamelion: Reliable change detection for long-term LiDAR mapping in transient environments," *IEEE Robotics and Automation Letters*, vol. 11, no. 4, pp. 4361–4368, 2026.

[187] R. Bai, H. Guo, W.-Y. Yau, and L. Xie, "Graph-based SLAM-aware exploration with prior topo-metric information," *IEEE Robotics and Automation Letters*, vol. 9, no. 9, pp. 7597–7604, 2024.

[188] J. Jiang, Y. Zhu, Z. Wu, and J. Song, "DualMap: Online open-vocabulary semantic mapping for natural language navigation in dynamic changing scenes," *IEEE Robotics and Automation Letters*, vol. 10, no. 12, pp. 12 612–12 619, 2025.

[189] A. H. Tan, A. Fung, H. Wang, and G. Nejat, "Mobile robot navigation using hand-drawn maps: A vision language model approach," *IEEE Robotics and Automation Letters*, vol. 10, no. 7, pp. 7667–7674, 2025.

[190] F. Xie, S. Schwertfeger, and H. Blum, "osmAG-LLM: Zero-shot open-vocabulary object navigation via semantic maps and large language models reasoning," *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 2426–2433, 2026.

[191] T. Rouček, Z. Rozsypálek, J. Blaha, J. Ulrich, and T. Krajník, "Predictive data acquisition for lifelong visual teach, repeat and learn," *IEEE Robotics and Automation Letters*, vol. 9, no. 11, pp. 10 042–10 049, 2024.

[192] D. Cattaneo and A. Valada, "CMRNext: Camera to LiDAR matching in the wild for localization and extrinsic calibration," *IEEE Transactions on Robotics*, vol. 41, pp. 1995–2013, 2025.

[193] A. Asgharivaskasi and N. Atanasov, "Semantic OcTree mapping and Shannon mutual information computation for robot exploration," *IEEE Transactions on Robotics*, vol. 39, no. 3, pp. 1910–1928, 2023.

[194] B. Liu, X. Xiao, and P. Stone, "A lifelong learning approach to mobile robot navigation," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 1090–1096, 2021.

[195] T. Soualhi, N. Crombez, A. Lombard, Y. Ruichek, and S. Galland, "Leveraging motion perceptibility and deep reinforcement learning for visual control of nonholonomic mobile robots," *Robotics and Autonomous Systems*, vol. 189, p. 104920, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0921889025000065

[196] B. Sharma, P. Jadhav, P. Paul, K. Krishna, and A. K. Singh, "MonoMPC: Monocular vision based navigation with learned collision model and risk-aware model predictive control," *IEEE Robotics and Automation Letters*, vol. 11, no. 2, pp. 1330–1337, 2026.

[197] D. Wang *et al.*, "G-DRAGON: Geospatial reasoning and dynamic planning for retrieval-augmented outdoor navigation," *IEEE Robotics and Automation Letters*, vol. 11, no. 7, pp. 8768–8775, 2026.

[198] V. S. Dorbala, J. F. Mullen, and D. Manocha, "Can an embodied agent find your "cat-shaped mug"? LLM-based zero-shot object navigation," *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 4083–4090, 2024.

[199] A. Saboury and M. K. Uyguroglu, "Uncertainty-aware real-time visual anomaly detection with conformal prediction in dynamic indoor environments," *IEEE Robotics and Automation Letters*, vol. 10, no. 5, pp. 4468–4475, 2025.

[200] N. Jadhav, M. Behari, R. J. Wood, and S. Gil, "WiSER-X: Wireless signal-based decentralized multi-robot exploration without explicit information exchange," *IEEE Robotics and Automation Letters*, vol. 11, no. 9, pp. 10 983–10 990, 2026.

[201] A. Athukorala, B. P. L. Lau, K. Pongsirijinda, C. Yuen, and U.-X. Tan, "Multi-robot collaborative SLAM (Multi-SLAM) with distributed lightweight predictive frontier exploration (LPFE)," *IEEE Robotics and Automation Letters*, vol. 11, no. 2, pp. 2274–2281, 2026.

[202] Q. Tang, H. Liu, M. Yu, and Y. Wang, "Parallel graph inference for decentralized and asynchronous multi-robot cooperative localization under limited connectivity," *IEEE Robotics and Automation Letters*, vol. 11, no. 10, pp. 12 120–12 127, 2026.

[203] X. Liu *et al.*, "SlideSLAM: Sparse, lightweight, decentralized metric-semantic SLAM for multirobot navigation," *IEEE Transactions on Robotics*, vol. 41, pp. 6529–6548, 2025.

[204] M. Luperto, M. Tellaroli, M. Antonazzi, and N. Basilico, "Multi-robot rendezvous in communication-restricted unknown environments via backtracking and semantic frontier-based exploration," *Robotics and Autonomous Systems*, vol. 194, p. 105137, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0921889025002349

[205] A. R. Da Silva, L. Chaimowicz, T. C. Silva, and M. A. Hsieh, "Communication-constrained multi-robot exploration with intermittent rendezvous," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 3490–3497.

[206] E. Latif and R. Parasuraman, "Communication-efficient multi-robot exploration using coverage-biased distributed Q-learning," *IEEE Robotics and Automation Letters*, vol. 9, no. 3, pp. 2622–2629, 2024.

[207] Z. Zhang, J. Yu, J. Tang, Y. Xu, and Y. Wang, "MR-TopoMap: Multi-robot exploration based on topological map in communication restricted environment," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 10 794–10 801, 2022.

[208] K. B. Naveed, D. R. Agrawal, R. Kumar, and D. Panagou, "Adaptive ergodic search with energy-aware scheduling for persistent multi-robot

missions," *Autonomous Robots*, vol. 49, no. 4, p. 27, Sep 2025. [Online]. Available: https://doi.org/10.1007/s10514-025-10215-6

[209] T.-S. Nguyen, H.-N. Cao, N. H. Thai, and M. T. Pham, "C-CBF: Communication-aware control barrier functions for resilient multi-robot connectivity," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 5596–5614, 2026.

[210] X. Zhang, J. Chen, Y. Zhu, B. Luo, and M. Guo, "CoCoPlan: Adaptive coordination and communication for multi-robot systems in dynamic and unknown environments," *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 3270–3277, 2026.

[211] G. Niu, S. Nuerdaolieti, J. Li, Z. Jing, L. Hou, and J. Sun, "Feedback-guided hierarchical transmission for multi-robot mapping under bandwidth constraints," *IEEE Robotics and Automation Letters*, vol. 11, no. 7, pp. 8576–8583, 2026.

[212] Q. Bi, X. Zhang, S. Zhang, Q. Dong, and J. Yuan, "URGE: Efficient decentralized multirobot exploration guided by unknown regions under limited communication," *IEEE Transactions on Robotics*, vol. 42, pp. 3028–3047, 2026.

[213] K. Pongsirijinda, Z. Cao, B. P. L. Lau, R. Liu, C. Yuen, and U.-X. Tan, "MEF-Explore: Communication-constrained multi-robot entropy-field-based exploration," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 16 062–16 078, 2025.

[214] C. Wang *et al.*, "Design and implementation of cloud–edge collaborative mapping and control system for mobile robots," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 11 096–11 116, 2026.

[215] X. Kong, Y. Xia, Z. Sun, D.-H. Zhai, Y. Deng, and S. Zhang, "Differential high order control barrier function-based safe reinforcement learning," *IEEE Robotics and Automation Letters*, vol. 10, no. 7, pp. 7524–7531, 2025.

[216] W. Wang and D. Meng, "Constrained predictive learning control for omnidirectional wheeled mobile robots via reducing iteration horizon," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 15 513–15 525, 2025.

[217] Y. Ma *et al.*, "Safety-critical control of nonholonomic vehicle trajectory tracking via risk-aware zone control barrier functions," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 13 469–13 479, 2026.

[218] Y. Qiao, X. He, A. Zhuo, Z. Sun, W. Bao, and Z. Li, "Curvature-constrained vector field for motion planning of nonholonomic robots," *IEEE Transactions on Robotics*, vol. 42, pp. 455–474, 2026.

[219] S. Yin, C. Chen, and Z. Xiang, "Predefined-time fault-tolerant formation control for large-scale heterogeneous wheeled mobile robot swarms," *IEEE Systems Journal*, vol. 20, no. 3, pp. 771–780, 2026.

[220] S. Rajkumar, C. Tiriolo, and W. Lucia, "A set-theoretic control strategy for a platoon of constrained differential-drive robots with inter-vehicle collision avoidance," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 4213–4222, 2026.

[221] Z. Du, H. Zhang, P. Cui, Z. Wang, and H. Yan, "Safe trajectory generation for nonholonomic multi-robot systems: A compensation-based MPC approach," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 21 831–21 842, 2025.

[222] H. Wu, S. Wang, H. Li, Y. Xie, S. Zheng, and S. Q. Xie, "Adaptive fault-tolerant control of wheeled mobile robots with multiple actuator faults and saturation," *IEEE Robotics and Automation Letters*, vol. 10, no. 4, pp. 4156–4163, 2025.

[223] J.-X. Zhang, J. Ding, and T. Chai, "Fault-tolerant prescribed performance control of wheeled mobile robots: A mixed-gain adaption approach," *IEEE Transactions on Automatic Control*, vol. 69, no. 8, pp. 5500–5507, 2024.

[224] W. Ao, H. Yin, S. Sun, P. Liu, and P. Shi, "Fixed-time fault-tolerant trajectory tracking control for a wheeled mobile robot," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 811–819, 2026.

[225] H. Guo, Q. Kang, W.-Y. Yau, C.-M. Chew, and D. Rus, "R-FAC: Resilient value function factorization for multirobot efficient search with individual failure probabilities," *IEEE Transactions on Robotics*, vol. 41, pp. 3385–3401, 2025.

[226] N. Kumar, J. J. Lee, S. Rathinam, S. Darbha, P. B. Sujit, and R. Raman, "The persistent robot charging problem for long-duration autonomy," *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2191–2198, 2025.

[227] S. Shahsavari, H. Haghbayan, A. Miele, E. Immonen, and J. Plosila, "A coordinated approach to control mechanical and computing resources in mobile robots," *IEEE Transactions on Robotics*, vol. 41, pp. 347–363, 2025.

[228] A. B. Asghar, S. Sundaram, and S. L. Smith, "Multirobot persistent monitoring: Minimizing latency and number of robots with recharging constraints," *IEEE Transactions on Robotics*, vol. 41, pp. 236–252, 2025.